

A Calibrated Bayesian Bootstrap Confidence Interval for Mokken's Scale-Level Scalability Coefficient

JoonHo Lee

ABSTRACT. Mokken scale analysis condenses the question of whether a fixed set of binary items forms a scale into a single scalability coefficient, and applied practice compares that coefficient with conventional benchmarks as if it were free of sampling error. Standard errors are available, but the symmetric intervals built from them assume a shape that this bounded ratio statistic does not have in the samples where it is actually used, and how often such intervals cover the true coefficient had not been established. We construct a confidence interval by repeatedly reweighting the observed respondents with the Bayesian bootstrap and then widening the result by a small deterministic factor. The widening corrects a measured shortfall in coverage, and its single constant was chosen on training data, validated on held-out data, and locked before the final evaluation existed. In a demanding confirmatory simulation the calibrated interval held its advertised coverage in every scenario family at a negligible cost in length, whereas our implementations of the standard formula-based interval and of two refined bootstrap corrections missed far more often and, for longer scales, frequently failed to produce an interval at all. An application to a public scientific-reasoning scale illustrates the method returning a stable and unfavorable answer. The procedure is implemented in the R package `bayesmokken`, and we suggest that a reported coefficient near a conventional benchmark should routinely carry such an interval.

Keywords: Mokken scale analysis; scalability coefficient; Bayesian bootstrap; confidence interval; nonparametric item response theory

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1. Introduction

Mokken scale analysis is widely used in psychological measurement to assess whether a fixed set of items orders respondents consistently enough to be scored together. The method is built on the monotone homogeneity model of nonparametric item response theory (Mokken, 1971; Sijtsma & Molenaar, 2002), and its central summary is Loevinger’s scalability coefficient H (Loevinger, 1948). Much of applied practice is organized around the scale-level value of this coefficient: textbooks and tutorials present the conventional bands ($H < .3$ unscalable, $.3$ to $.4$ weak, $.4$ to $.5$ moderate, $\geq .5$ strong), automated item selection procedures accept and reject items against a lower bound for H , and applied articles in clinical, educational, and personality research report the coefficient as the principal evidence that an item set forms a scale (Mokken, 1971; Sijtsma & van der Ark, 2017; Stochl et al., 2012; Watson et al., 2012).

The inferential support for this practice is weaker than is commonly appreciated. A point estimate of H is a sample statistic, yet it is often interpreted as if it were the population value: a scale whose estimated coefficient is $.32$ is called scalable and one at $.28$ is not, although sampling variability alone can account for the difference. The methodological literature has addressed part of the problem. Delta-method standard errors for scalability coefficients are available and implemented (Kuijpers et al., 2013; van der Ark, 2007), with extensions to two-level designs (Koopman, Zijlstra, & van der Ark, 2020; Koopman et al., 2022) and careful studies of their small-sample bias (Kuijpers et al., 2016). However, an accurate standard error does not by itself provide a calibrated interval. Wald-type intervals constructed from analytic standard errors inherit the symmetry and normality of their construction rather than the finite-sample behavior of a bounded ratio statistic, and, as our results indicate, coverage can deteriorate in the conditions where applied users typically work, namely modest samples, scales of moderate length, and coefficients near the conventional decision bands.

In this article we develop, validate, and apply a different approach to interval estimation for the fixed binary scale-level coefficient. The starting point is Rubin’s (1981) Bayesian bootstrap (BB): respondents are reweighted by Dirichlet random weights, each reweighted empirical distribution induces a value of the scalability functional, and the equal-tail quantiles of the resulting distribution of H values form an interval. The BB is attractive for this functional for several reasons. It requires no derivative

formulas and no asymptotic plug-in; it evaluates the ratio form of the statistic directly on each reweighted distribution; its smooth weighting avoids the discreteness of resampling with replacement in small samples; and it possesses a first-order theory that we make explicit for this functional. Two further steps are taken beyond this starting point. First, we show that the raw BB interval, like its competitors, under-covers slightly in finite samples, and we correct it by the smallest modification we could justify: the interval is widened about its midpoint by a deterministic factor that exceeds one by a fixed constant divided by the square root of the sample size. The correction moves the endpoints by less than the order of the interval width itself, so it provably preserves the BB interval’s first-order justification, and its constant was selected on training conditions, evaluated once on held-out conditions, and then confirmed prospectively, so that it was never tuned on the data that evaluate it. Second, we subject the locked method to a fresh confirmatory simulation spanning four data-generating families constructed to strain the statistic in different ways, with results reported under a prespecified rule: unconditional coverage, with undefined intervals counted as misses.

In the confirmatory test the calibrated interval’s family-level coverage ranged from 94.5% to 95.1% at the nominal 95% level, at the cost of increasing mean interval length by less than 3% over the raw BB interval. Under the same protocol, an influence-function delta-method Wald interval and a BB-based bias-corrected accelerated construction, both implemented under a strict regularity screen whose refusals count as misses, covered between 82.6% and 91.8% and as little as 71.4%, respectively. We then apply the frozen method, without retuning, to a real fixed scale, the 11-item Scientific Reasoning Scale (Drummond & Fischhoff, 2017) in an Italian administration (Caliciuri & Lanz, 2024), where it produces a stable interval lying entirely below the conventional .30 reference; this negative empirical finding is reported as such. The complete procedure is implemented in the open-source R package `bayesmokken`.

The article makes four contributions. First, it gives the fixed binary scale-level Mokken coefficient an explicit estimand-first treatment: a population functional, its influence function, and conditions under which BB inference matches first-order sampling theory. Second, it introduces a minimal width calibration with exact finite-sample properties (the calibrated interval always contains the raw interval and cannot reduce coverage), together with a proof that the correction leaves the first-order limit unchanged. Third, it reports a validation design that is unusually strict for this literature, with training, held-out selection, public locking, and a fresh confirmatory

test under blocking criteria, so that the reported coverage constitutes a prediction evaluated on untouched data rather than a selected summary. Fourth, it provides an application, software, and a complete reproducibility chain. Throughout, the interval is presented as a BB-based confidence interval with repeated-sampling semantics, and the evidence concerns fixed binary complete-response scales for independent respondents; Section 7 states the boundaries in full.

The plan of the article is as follows. In Section 2 we review scalability coefficients and the existing uncertainty machinery. In Section 3 we define the estimand, establish first-order properties of the estimator and of the BB, and develop the width calibration together with its exact finite-sample and asymptotic properties. In Section 4 we state the exact algorithm, describe how the constant was fixed, and document the software. Sections 5 and 6 report the confirmatory simulation and the empirical illustration, and Section 7 concludes. Proofs, exact algorithm and comparator specifications, data-generating details, the governance chronology, the complete confirmatory tables, and the application details are collected in the online supplement, Appendices A–H of this document.

2. Background: Scalability Coefficients and Their Uncertainty

2.1. The Monotone Homogeneity Model and Loevinger’s H . Mokken’s (1971) monotone homogeneity model (MHM) assumes a unidimensional latent variable, conditional independence of item responses given that variable, and item response functions that are nondecreasing in it. Notably, the model assumes neither a logistic form nor a parametric latent distribution. Its observable implications are nonnegative-association constraints (conditional association in the sense of Holland and Rosenbaum, 1986, with refinements by Junker, 1993, Junker and Ellis, 1997, and Junker and Sijtsma, 2000), and its principal applied instrument is the family of scalability coefficients.

For binary items i and j with success probabilities p_i and p_j , the pair coefficient is the covariance of the two items divided by its maximum value attainable under the fixed margins,

$$H_{ij} = \frac{\text{Cov}(X_i, X_j)}{\text{Cov}^{\max}(X_i, X_j)} = \frac{m_{ij} - p_i p_j}{\min(p_i, p_j) - p_i p_j}, \quad (1)$$

where $m_{ij} = \mathbb{E}(X_i X_j)$. Item and scale coefficients aggregate numerators and denominators before dividing, so that the scale coefficient is not the average of the pair coefficients, and the scale-level version admits the equivalent Guttman-error form $H = 1 - F/E_0$, the observed count of Guttman errors relative to its expectation

under item independence (Loevinger, 1948; Mokken, 1971). Under the MHM every H_{ij} is nonnegative, and $H = 1$ exactly when the data contain no Guttman errors; between these extremes, H measures how strongly the item set orders respondents relative to the maximum that its margins allow.

This interpretation explains the central role of H in applied Mokken analysis. Mokken’s rules of thumb ($H \geq .3$ for a scale at all, with .4 and .5 marking moderate and strong scalability) organize applied reporting (Sijtsma & van der Ark, 2017; Sijtsma et al., 2011), drive the automated item selection procedure (Straat et al., 2013; van Abswoude et al., 2004), and anchor recommendations in clinical and educational applications (Stochl et al., 2012; Watson et al., 2012; Wind, 2017). The same literature has warned that coefficient-driven selection can be misleading about dimensionality (Crişan et al., 2022; Smits et al., 2012); such warnings reinforce, rather than remove, the need to treat any reported coefficient as an estimate.

2.2. Existing Uncertainty Machinery. The sampling behavior of estimated scalability coefficients has been studied from several directions. van Onna (2004) explored computer-intensive confidence intervals for H , advocating a range-preserving nonparametric bootstrap interval. The major analytic advance is the delta-method standard errors of Kuijpers et al. (2013), derived in a marginal-modeling framework (van der Ark et al., 2008) and implemented in the `mokken` package (van der Ark, 2007); subsequent work documented small-sample bias in both the point estimates and the standard errors (Kuijpers et al., 2016) and extended the machinery to two-level designs (Koopman, Zijlstra, de Rooij, & van der Ark, 2020; Koopman, Zijlstra, & van der Ark, 2020; Koopman et al., 2022, 2023). Sample size guidance for Mokken analysis has likewise been developed with confidence interval coverage as one criterion (Straat et al., 2014).

However, an accurate standard error becomes an interval only through a construction, and the default construction, the symmetric Wald interval $\hat{H} \pm 1.96 \widehat{SE}$, imports assumptions that the statistic does not satisfy in small samples. The estimator of Equation (1)’s aggregate is a ratio of correlated sums; its finite-sample distribution is skewed, its variance depends on the margins, and near the boundaries (margins close to ties, denominators close to zero) the normal approximation deteriorates. Resampling offers an alternative, since percentile, basic, studentized, and bias-corrected accelerated (BCa) intervals are all available in principle (Beran, 1987; DiCiccio & Efron, 1996). However, resampling with replacement produces discrete weight configurations that can degenerate at small n , and, as our confirmatory study shows

for strict-regularity-screened implementations of these constructions, the more elaborate variants (studentized, BCa) are the ones that lose definedness most severely for this functional. What is lacking for the applied user is therefore not an uncertainty formula but an interval whose finite-sample coverage has been demonstrated, under demanding conditions, for the coefficient that is actually reported.

2.3. The Bayesian Bootstrap. The Bayesian bootstrap (Rubin, 1981) replaces resampling with reweighting: given observed response vectors $X_1, \dots, X_n$, draw $(W_1, \dots, W_n) \sim \text{Dirichlet}(1, \dots, 1)$ and form the reweighted empirical distribution $P_n^W = \sum_r W_r \delta_{X_r}$. Each draw induces a value $H(P_n^W)$ of any functional of the response distribution. Large-sample theory for the BB is classical: its conditional distribution matches the first-order sampling distribution of the empirical functional (Lo, 1987), with finite-population versions (Lo, 1988) and likelihood-weighted generalizations (Lyddon et al., 2019; Newton & Raftery, 1994; Newton et al., 2021) extending its reach; Chamberlain and Imbens (2003) treat it as a general tool for nonparametric estimands, and recent work applies hierarchical variants to causal estimands (Oganisian et al., 2024).

Three properties make the BB well suited to scalability coefficients. First, it is derivative-free: whereas the delta method must differentiate the ratio $H = A/B$, including its min-kinks, the BB simply recomputes the functional under smooth weight perturbations. Second, every draw evaluates the same functional form on a legitimate reweighted distribution, so each $H(P_n^W)$ automatically respects the algebraic structure of the coefficient, including the possibility that the population coefficient lies near a boundary where symmetric intervals perform poorly. Third, the flat Dirichlet weights are strictly positive, so no respondent is ever deleted, which keeps small- n behavior smooth where multinomial resampling becomes discrete.

Although the BB has a well-known reading as a noninformative posterior over distributions supported on the observed data, we use it purely as a device for approximating repeated-sampling uncertainty: every interval in this article is evaluated by, and reported with, frequentist coverage semantics, and we therefore refer to the result as a BB-based confidence interval and never as a credible interval. This choice also separates our aim from the Bayesian and Bayesian nonparametric literatures on monotone item response functions and model checking, which include order-constrained inference (Karabatsos & Sheu, 2004; van Onna, 2002), Bayes-factor tests of monotonicity (Tijmstra & Bolsinova, 2019; Tijmstra et al., 2015), and flexible response-function

models (Arenson & Karabatsos, 2018; Duck-Mayr et al., 2020; Duncan & MacEachern, 2008; Karabatsos, 2015; Wang & Kingston, 2020). Those literatures model the response process, whereas our target is calibrated uncertainty for the summary coefficient that practitioners already report.

3. Estimand and Theory

3.1. The Population Coefficient as an Estimand. Let $X = (X_1, \dots, X_J) \in \{0, 1\}^J$ be the response vector for a *fixed, prespecified* set of J binary items, distributed according to an unknown population law P , and write $p_i(P) = \mathbb{E}_P X_i$ and $m_{ij}(P) = \mathbb{E}_P(X_i X_j)$. Define, for each pair $i < j$,

$$C_{ij}(P) = m_{ij}(P) - p_i(P)p_j(P), \quad D_{ij}(P) = \min\{p_i(P), p_j(P)\} - p_i(P)p_j(P), \quad (2)$$

and the aggregates $A(P) = \sum_{i < j} C_{ij}(P)$ and $B(P) = \sum_{i < j} D_{ij}(P)$. The estimand of this article is the population scale coefficient

$$H(P) = \frac{A(P)}{B(P)}, \quad B(P) > 0. \quad (3)$$

The estimator is the plug-in $\hat{H} = H(P_n)$, with P_n the empirical distribution of n independent, complete response vectors. Three restrictions define the scope of every claim that follows: the item set is fixed in advance (no selection, deletion, or reversal), the items are binary, and respondents are independent with complete responses. Here D_{ij} is the maximum covariance compatible with the margins, the comonotone bound of Rüschendorf (1982) (see also Puccetti and Wang, 2015), so that Equation (3) is a margin-normalized association measure rather than a raw covariance.

Definition 3.1 (Regular domain). P lies in the regular domain if (a) all item margins are interior, $0 < p_i(P) < 1$; (b) the margins have a strict ordering (no exact ties), so that each min in Equation (2) is attained by a unique item (one active-min branch); (c) $B(P)$ is bounded away from zero; and (d) the influence-function variance σ_H^2 in Equation (4) below is positive and finite.

Smooth large-sample theory applies on the regular domain. Exact marginal ties switch the active branch of the min, whereas margins at 0 or 1 and denominators at zero remove differentiability altogether. Condition (b) deserves particular attention, since it is the one that tightens as scales grow: with many items the margins crowd the unit interval, near-ties become common, and any construction that must differentiate across the min-kinks inherits the difficulty. This observation anticipates the

comparator failures reported in [Section 5](#). Our theoretical statements hold on regular cells, the simulation separates regular from deliberately irregular (stress) conditions, and boundary behavior is examined empirically rather than assumed to be absent. Directional analyses of such kinked functionals exist (Fang & Santos, 2019) but are not needed for the claims we make.

3.2. First-Order Behavior of $\hat{H}$ and of the Bayesian Bootstrap. On the regular domain, each C_{ij} and D_{ij} is a smooth functional of the finite-dimensional moment vector, and the quotient in [Equation \(3\)](#) linearizes. Writing ϕ_A and ϕ_B for the influence functions of A and B (explicit forms in [Appendix B](#)), the influence function of H is the quotient form

$$\psi_H(y; P) = \frac{\phi_A(y; P) - H(P) \phi_B(y; P)}{B(P)}, \quad (4)$$

and the estimator is asymptotically linear.

Proposition 3.1 (Asymptotic linearity). *If P lies in the regular domain of [Definition 3.1](#), then $\mathbb{E}_P \psi_H(Y; P) = 0$, $H(P_n) - H(P) = n^{-1} \sum_{r=1}^n \psi_H(X_r; P) + o_p(n^{-1/2})$, and $\sqrt{n}\{H(P_n) - H(P)\} \xrightarrow{d} \mathcal{N}(0, \sigma_H^2)$ with $\sigma_H^2 = \text{Var}_P\{\psi_H(Y; P)\}$.*

The same linearization governs the Bayesian bootstrap. Let $(W_1, \dots, W_n) \sim \text{Dirichlet}(1, \dots, 1)$ independently of the data and $P_n^W = \sum_r W_r \delta_{X_r}$.

Proposition 3.2 (First-order BB matching). *Under the conditions of [Proposition 3.1](#), the conditional law of $\sqrt{n}\{H(P_n^W) - H(P_n)\}$ given $X_{1:n}$ converges in probability to $\mathcal{N}(0, \sigma_H^2)$, the same limit as the sampling law of $\sqrt{n}\{H(P_n) - H(P)\}$.*

[Proposition 3.2](#) justifies the raw BB interval: take B weight draws, evaluate $H(P_n^W)$ on each, and report the equal-tail 2.5% and 97.5% quantiles $[L_n, U_n]$. The justification is, however, first order only. The proposition guarantees that the interval's coverage converges to the nominal level; it says nothing about the rate of convergence, and it does not apply outside the regular domain. Both derivative formulas underlying [Equation \(4\)](#) and the linearization itself were verified numerically against independent oracles across 1,561 checks with zero failures (maximum absolute derivative error below 1.6×10^{-8} ; details in [Appendix B](#)).

3.3. Local Curvature and the Order of a Finite-Sample Correction. If first-order theory were exact, the raw BB interval would require no correction. However, $H = A/B$ is a curved functional, and at realistic sample sizes the curvature is not

negligible. Within a regular active-min cell the second derivative of the quotient along zero-mass directions Q, R is

$$H_P''[Q, R] = \frac{A_P''[Q, R]}{B} - \frac{A_P'[Q] B_P'[R] + A_P'[R] B_P'[Q] + A B_P''[Q, R]}{B^2} + \frac{2A B_P'[Q] B_P'[R]}{B^3}, \quad (5)$$

and the Dirichlet covariance structure yields an exact identity for the conditional expectation of the local quadratic term: with $z_r = \delta_{X_r} - P_n$ and $\hat{\kappa}_r = H_{P_n}''[z_r, z_r]$,

$$\mathbb{E}_W\left\{\frac{1}{2}H_{P_n}''[P_n^W - P_n, P_n^W - P_n]\right\} = \frac{1}{2n(n+1)} \sum_{r=1}^n \hat{\kappa}_r. \quad (6)$$

The identity in [Equation \(6\)](#) (verified numerically to 2.4×10^{-17}) places the leading nonlinear effect of Dirichlet reweighting at exactly the $O_p(n^{-1})$ scale. We draw a deliberately limited conclusion from it: any finite-sample correction of the raw interval should be of order n^{-1} . At that order we adopt the simplest candidate, a midpoint-symmetric width adjustment, as a design choice; the identity constrains the scale of the correction but not its shape. The curvature analysis therefore motivates the order of the correction below and nothing more, since it fixes neither the form, nor the full BB mean shift, nor a coverage direction, nor the value of the constant. That the raw interval tends to be too short, so that the appropriate correction is a widening, is an empirical finding of the frozen training stage that was subsequently confirmed; it is not deduced here.

3.4. The Minimal Width Calibration. Let $[L_n, U_n]$ be a defined raw equal-tail BB interval with midpoint $m_n = (L_n + U_n)/2$ and half-width $w_n = (U_n - L_n)/2$. The calibrated interval multiplies the half-width by a deterministic factor,

$$[\tilde{L}_n, \tilde{U}_n] = [m_n - k_n w_n, m_n + k_n w_n], \quad k_n = 1 + \frac{\gamma}{\sqrt{n}}, \quad \gamma = .25. \quad (7)$$

No endpoint clipping is applied, and an undefined raw interval remains undefined; the calibration never repairs a failed base interval. Three facts about this map are exact, not asymptotic.

Proposition 3.3 (Exact finite-sample properties). *For any $n \geq 1$, $\gamma \geq 0$, and defined raw interval: (a) the midpoint is unchanged; (b) $[L_n, U_n] \subseteq [\tilde{L}_n, \tilde{U}_n]$; (c) the length is multiplied by exactly k_n ; and (d) for every fixed θ , $\mathbf{1}\{\theta \in [L_n, U_n]\} \leq \mathbf{1}\{\theta \in [\tilde{L}_n, \tilde{U}_n]\}$, so on any set of replications the calibrated interval's coverage is at least the raw interval's.*

Lemma 3.1 (The perturbation is $O_p(n^{-1})$). *If γ is a fixed constant and the raw half-width satisfies $w_n = O_p(n^{-1/2})$ (as it does on the regular domain by [Proposition 3.2](#)), then $\tilde{L}_n - L_n = -\gamma w_n / \sqrt{n} = O_p(n^{-1})$ and likewise for the upper endpoint; in particular $\sqrt{n}(\tilde{L}_n - L_n) = o_p(1)$ and $\sqrt{n}(\tilde{U}_n - U_n) = o_p(1)$.*

Corollary 3.1 (First-order coverage limit preserved). *Suppose in addition that $(\sqrt{n}\{L_n - \theta\}, \sqrt{n}\{U_n - \theta\})$ converges jointly in distribution with a limit law placing no mass on the coverage boundary. Then the calibrated endpoints have the same joint limit, and $\mathbb{P}\{L_n \leq \theta \leq U_n\} - \mathbb{P}\{\tilde{L}_n \leq \theta \leq \tilde{U}_n\} \rightarrow 0$.*

Taken together, [Proposition 3.3–3.1](#) confine the correction within narrow and verifiable limits: in every finite sample it can only add coverage, its absolute endpoint change is $O_p(n^{-1})$ with a relative length cost of exactly $\gamma/\sqrt{n}$, and asymptotically it leaves the first-order properties established in [Proposition 3.2](#) unchanged. Because [Proposition 3.3\(d\)](#) holds replication by replication, it holds under any data-generating process whatever, so the question the simulation must settle is not whether the calibration can harm coverage but how much it helps and at what cost in length. What these results do not supply is a second-order theory: we prove no Edgeworth expansion and no coverage-error theorem, and no optimality is claimed for $\gamma = .25$, which the next two sections fix and evaluate empirically. Proofs of all statements appear in [Appendix B](#).

4. The Calibrated Interval Procedure

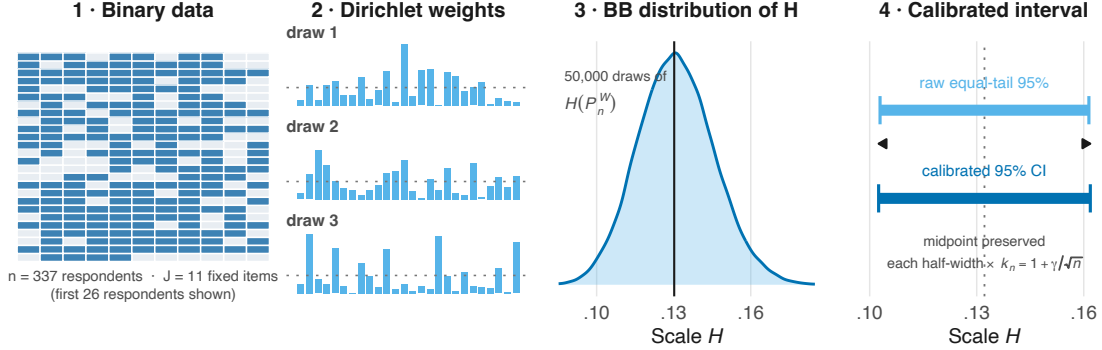
4.1. Exact Algorithm. For a fixed binary item set with n complete respondents, confidence level 95%, and B weight draws, the procedure is:

- (1) *Weights.* Draw B independent respondent weight vectors $(W_1, \dots, W_n) \sim \text{Dirichlet}(1, \dots, 1)$ from a seeded, draw-major stream.
- (2) *Functional.* For each draw, evaluate the plug-in coefficient $H(P_n^W)$ of [Equation \(3\)](#) on the reweighted distribution (the identical ratio-of-sums form used for the point estimate, computed on weighted margins and weighted joint moments). A draw whose weighted denominator is zero is recorded as undefined, not discarded and not imputed.
- (3) *Raw interval.* If at least the required number of draws is defined, set $[L_n, U_n]$ to the type-8 (median-unbiased) empirical quantiles of the defined draws at 2.5% and 97.5%, as implemented in R’s `quantile` (R Core Team, [2026](#)).

- (4) *Calibration.* Report $[\tilde{L}_n, \tilde{U}_n] = [m_n - k_n w_n, m_n + k_n w_n]$ with $k_n = 1 + .25/\sqrt{n}$, per Equation (7).

Figure 1 illustrates the four steps with the application data of Section 6.

FIGURE 1. The Calibrated Bayesian Bootstrap Interval, End to End



Note. Illustration with the Scientific Reasoning Scale application data ($n = 337$, $J = 11$; Section 6). (1) The observed binary responses are fixed. (2) Respondent weights are drawn from a flat Dirichlet distribution; three illustrative draws are shown with the equal-weight line dotted. (3) Each weight vector induces a value of the scale coefficient $H(P_n^W)$; 50,000 draws form the BB distribution (vertical line: the sample estimate). (4) The raw equal-tail 95% interval is widened about its fixed midpoint by the deterministic factor $k_n = 1 + \gamma/\sqrt{n}$ with $\gamma = .25$, giving the calibrated interval. BB = Bayesian bootstrap; CI = confidence interval.

Failure semantics are inherited, typed, and strict: if the raw interval is undefined (insufficient defined draws, nonfinite endpoints, or a reversed interval), the calibrated interval is undefined as well, and in every performance summary of this article an undefined interval counts as a miss, so that coverage denominators are always all scheduled replications rather than the defined subset. No endpoint clipping to $[0, 1]$ or any other range is performed; a calibrated endpoint may therefore in principle fall outside the plausible range of H , but clipping would silently alter the coverage semantics of the reported interval. Per-replication method switching (“cascading” to another interval when one fails) is not permitted. The simulation and the empirical application use exactly this algorithm with $B = 2,000$ and $B = 50,000$, respectively; the confidence level is fixed at 95%, the only level for which calibration evidence exists.

4.2. Selection of the Constant $\gamma = .25$. The constant was fixed by a frozen train–validate–lock sequence, with each stage completed and locked before the next produced outcomes (Figure 2c).

Training. A candidate library of interval constructions, among them width-corrected variants of the raw equal-tail and basic BB intervals with γ on the preregistered grid $\{0, .25, .5, .75, 1, 1.5, 2, 2.5\}$, was evaluated on training cells (the SF01 subfamilies of the four data-generating families described in the next section). The grid, the candidates, and the acceptance criteria were frozen before any training outcome existed.

Held-out selection. Candidates surviving training were evaluated once on disjoint held-out validation cells (the SF02 subfamilies) against frozen eligibility criteria: family-level coverage inside the equivalence band $[.925, .975]$, protection against catastrophic cell-level undercoverage, definedness, and length efficiency. Exactly two primary candidates were eligible (the width-corrected raw equal-tail interval with $\gamma = .25$ and the width-corrected basic interval with $\gamma = 1.0$), and the prespecified selection rule (minimum equal-family, equal-cell conditional mean length) chose the former, whose intervals were 7.4% shorter; the prespecified tie rule was not needed. No parameter was changed after validation outcomes were seen.

Lock. The complete public method (estimator, weight law, B requirements, quantile type, $\gamma = .25$, no-clipping policy, failure semantics, and the confirmatory acceptance gates) was serialized into a public semantics lock before any confirmatory data were generated. The confirmatory test in the next section evaluates that frozen object; nothing about the method, the gates, or the analysis plan was altered afterward, and no confirmatory outcome fed back into any tuning decision.

We emphasize the status of the constant: $\gamma = .25$ is a simulation-calibrated quantity, selected in development, independently validated, and prospectively confirmed for fixed binary scale-level Mokken H . It is not analytically derived, and no optimality is claimed.

4.3. Software. The procedure is implemented in the open-source R package `bayesmokken` (Lee, 2026). The primary call returns the calibrated interval by default, with the raw equal-tail interval available as the reproducibility comparator:

```
fit <- bb_mokken_h(data, draws = 2000, seed = 1,
                  interval = "calibrated", level = 0.95)
fit$interval
```

The returned object records the method identifier, point estimate, raw and calibrated endpoints, γ , the width multiplier, defined and total draw counts, and a typed status. Within this binary path the package deliberately exposes no user-tunable γ

and no confidence level other than 95%, and undefined states are returned as typed statuses rather than repaired; the application data ship with the package as the `srs` dataset. The package also implements a second, separately developed and separately validated procedure for clustered ordinal scales, which uses a different estimand, a different weight law, and its own tuning constant; none of the evidence in this article bears on it, and the package refuses to cross the two paths. [Appendix H](#) records the package-level evidence for the binary path, the scope boundary between the two procedures, and the release history.

4.4. Transparency and Openness. We report how every simulation condition, Monte Carlo budget, and analysis rule was determined; there were no data exclusions, and no analyses beyond those in the frozen protocols were run. The simulation follows the reporting framework of Morris et al. (2019), and Monte Carlo uncertainty is reported with every coverage estimate (Koehler et al., 2009). The study was not preregistered in an external registry; instead, every stage was serialized before its outcomes existed into hash-locked protocol and receipt files that are archived, with their SHA-256 digests and the scripts that verify them, in the replication package ([Appendix E](#)). The empirical data are public: the Scientific Reasoning Scale administration of Caliciuri and Lanz (2024) is distributed under a CC-BY 4.0 license (OSF project `jk9dp`), and our analysis consumed a checksummed local mirror without modification. Analyses used R version 4.6.0 (R Core Team, 2026); point-estimate concordance was checked against `mokken` (van der Ark, 2007). All code, locked artifacts, and figure/table pipelines are available in the replication archive at <https://github.com/joonho112/bayesmokken-scalability-replication>, and the method itself is implemented in the `bayesmokken` package at <https://github.com/joonho112/bayesmokken> (Lee, 2026).

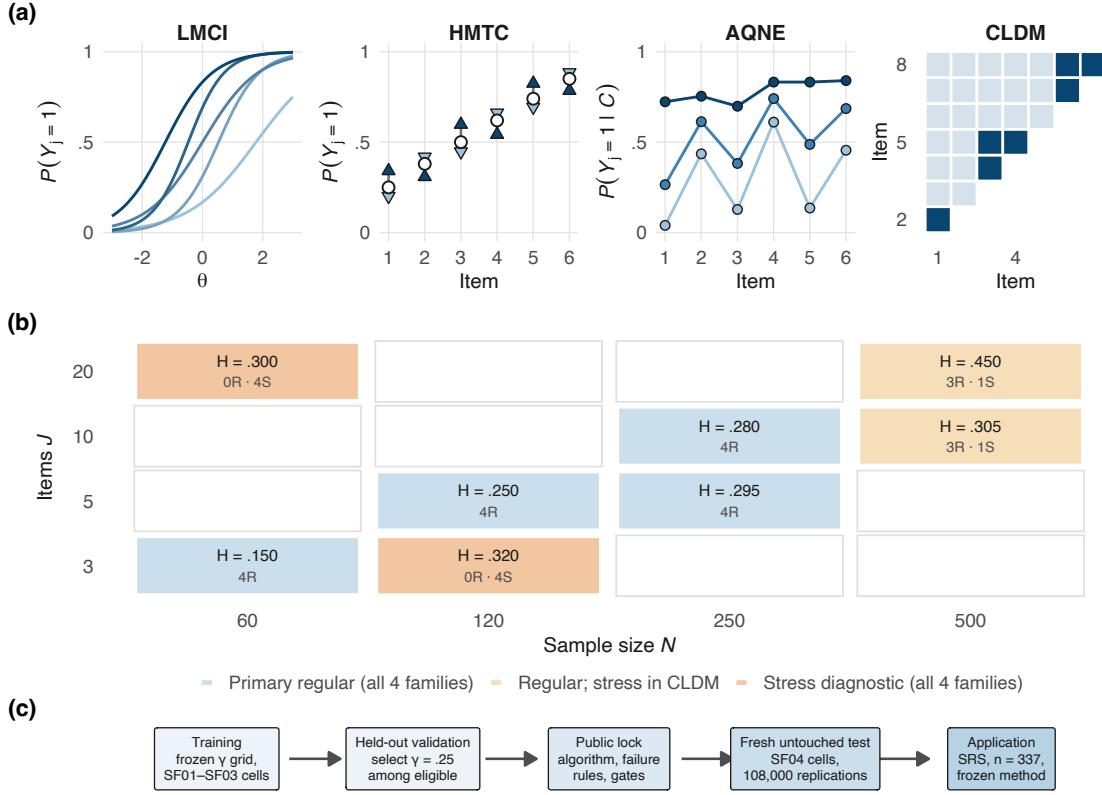
5. Confirmatory Simulation

The purpose of this study is confirmatory rather than exploratory: it evaluates the single frozen procedure of [Equation \(7\)](#), selected and locked before these data existed, against prespecified acceptance criteria on data-generating processes that played no role in development.

5.1. Design.

5.1.1. Data-Generating Families. Four DGP families probe the estimand from complementary directions ([Figure 2a](#); construction details and truth computations in

FIGURE 2. The Confirmatory Evaluation Design at a Glance



Note. (a) The four frozen data-generating mechanisms (illustrative parameter settings; the locked cell parameters live in the registries archived with the replication package). LMCI: monotone logistic response curves of a normal latent variable. HMTc: two latent classes (triangles) whose mixture holds each item margin exactly at its target (open circles). AQNE: class-conditional response profiles with item-specific, nonexchangeable quality. CLDM: a global class signal plus local residual dependence in item blocks (dark cells). (b) The eight $N \times J$ design points shared by all families, each with its population H target; fill shows how many of the four family versions of a cell are primary regular (R) versus stress diagnostic (S). (c) The frozen chronology: training and held-out selection on earlier subfamilies, a public method lock, the fresh untouched confirmatory test, and the empirical application, in that order and without feedback.

Appendix D). Each family is built to answer a different question about the interval. LMCI (latent monotone conditional independence) is the model-conforming benchmark: a standard-normal latent variable with monotone logistic response functions of item-specific location and slope, so that the data satisfy the monotone homogeneity model. It asks whether the interval performs as intended when the measurement model behind the coefficient is exactly right. HMTc (heterogeneous-margin target control) generates responses from two latent classes with item margins held exactly at

heterogeneous targets. Since the min denominators of Equation (2) are determined entirely by the margins, this family manipulates the raw material of the denominators directly rather than leaving it to chance; it asks whether coverage survives when the margin geometry is controlled adversarially. *AQNE* (asymmetric-quality nonexchangeable items) uses finite latent classes with item-specific locations and discriminations, breaking item exchangeability; it asks whether the interval depends on an incidental symmetry among items that the benchmark family happens to possess. *CLDM* (controlled local dependence) adds block-local residual factors to a global class signal, violating the auxiliary conditional-independence assumption while leaving the fixed-scale estimand well defined; it asks whether the interval remains valid for the functional when the measurement model behind it is violated in a controlled way. In every family the population coefficient is solved to a target value with audited numerical truth (maximum absolute truth residual 6.1×10^{-11} across the 32 cells).

5.1.2. *Cells, Budgets, and Lanes.* Each family contributes the same eight (N, J) design points with population H targets between .15 and .45 (Figure 2b): $N \in \{60, 120, 250, 500\}$ and $J \in \{3, 5, 10, 20\}$, spanning short screening scales to long inventories at applied sample sizes, with several targets placed near the conventional .30 band where uncertainty matters most. Of the resulting 32 cells, 22 satisfy the frozen regularity thresholds (interior margins, minimum margin-order gap, minimum pair and scale denominators) and form the *primary regular* lane on which the acceptance gates operate; the remaining 10 are deliberate *stress* cells (very small samples with many items, near-tied or near-degenerate denominators, and strong local dependence), which are reported separately and never combined with the primary claim. Regular cells received 4,000 scheduled replications and stress cells 2,000, for 108,000 replications per interval method, each with $B = 2,000$ BB draws from a deterministic seed stream. These budgets were fixed in the protocol; at 4,000 replications the Monte Carlo standard error of a cell-level coverage proportion near .95 is about .0035, and family-level MCSEs (reported throughout) are at most .0022. Because the four families were built for earlier phases of the frozen chain, the confirmatory run used freshly generated subfamily variants (structural profiles never used in training or validation) with new seeds.

5.1.3. *Comparators.* Six comparator intervals were evaluated under identical replications, draws, and failure accounting: the raw equal-tail BB interval (the base method of Equation (7)); basic, studentized, and BCa constructions computed from the same

BB draw stream (DiCiccio & Efron, 1996); the nonparametric bootstrap (NPB) percentile interval with multinomial resampling; and a delta-method Wald interval built from the influence-function standard error, in the spirit of analytic-SE practice (Kuijpers et al., 2013). One implementation policy should be noted at the outset. Our Wald, studentized, and BCa variants enforce the same strict regularity screen as the theory: an exact empirical margin tie, an empty active-min branch, or a degenerate transform yields a typed undefined interval, which unconditional coverage counts as a miss. Fielded software can resolve such states differently (for example, *mokken*’s analytic standard error averages the admissible tied orderings and returns a finite value), so these columns characterize the strict-screened variants specified in [Appendix C](#), not the behavior of any released package; we refer to them as the *screened* variants throughout.

5.1.4. *Prespecified Gates*. Nine blocking criteria, frozen with the method, define success: an integrity criterion (bit-level reproduction of the evaluation chain); family-level unconditional coverage inside the equivalence band $[\text{.925}, \text{.975}]$ for all four families; protection against catastrophic cell-level undercoverage; zero numerical-failure tolerance; a definedness criterion; two length-efficiency criteria against same-gate comparators; and two stress-lane criteria. Coverage is always unconditional, with undefined intervals counted as misses, so that a method cannot improve its coverage by failing selectively.

One further fact about the record should be reported. The confirmatory run below is the second scheduled attempt: the first, on a different fresh subfamily set, was terminated under the frozen fail-closed, no-retry rule after two infrastructure failures in its consolidation stage, before any metric, summary, or gate had been computed from its output, and that output was sealed. The full incident chronology, with hashes, is in [Appendix E](#).

5.2. Results.

5.2.1. *Primary Coverage*. All nine blocking gates passed. [Figure 3](#) and [Table 1](#) give the family-level picture. The calibrated interval’s unconditional coverage was .9491 (LMCI), .9506 (HMTc), .9484 (AQNE), and .9453 (CLDM), a range of 94.5% to 95.1%, inside the equivalence band in every family, including the local-dependence family whose generating model violates conditional independence. The raw equal-tail BB interval came close (.9404 to .9459) but was uniformly below nominal; the

calibration recovered +0.44 to +0.49 percentage points of coverage (the small finite-sample deficit, at the n^{-1} scale suggested by the curvature analysis, that the frozen training stage had identified), at a mean-length cost of 2.46% to 2.65% (Figure 3b). Only three methods passed every blocking criterion: the calibrated interval, its raw base, and the NPB percentile interval. Relative to the latter, the only comparator with comparable coverage, the calibrated interval was 1.5% to 2.5% shorter at equal or better family coverage, so the calibration is not achieved at a length cost that a standard resampling interval would avoid.

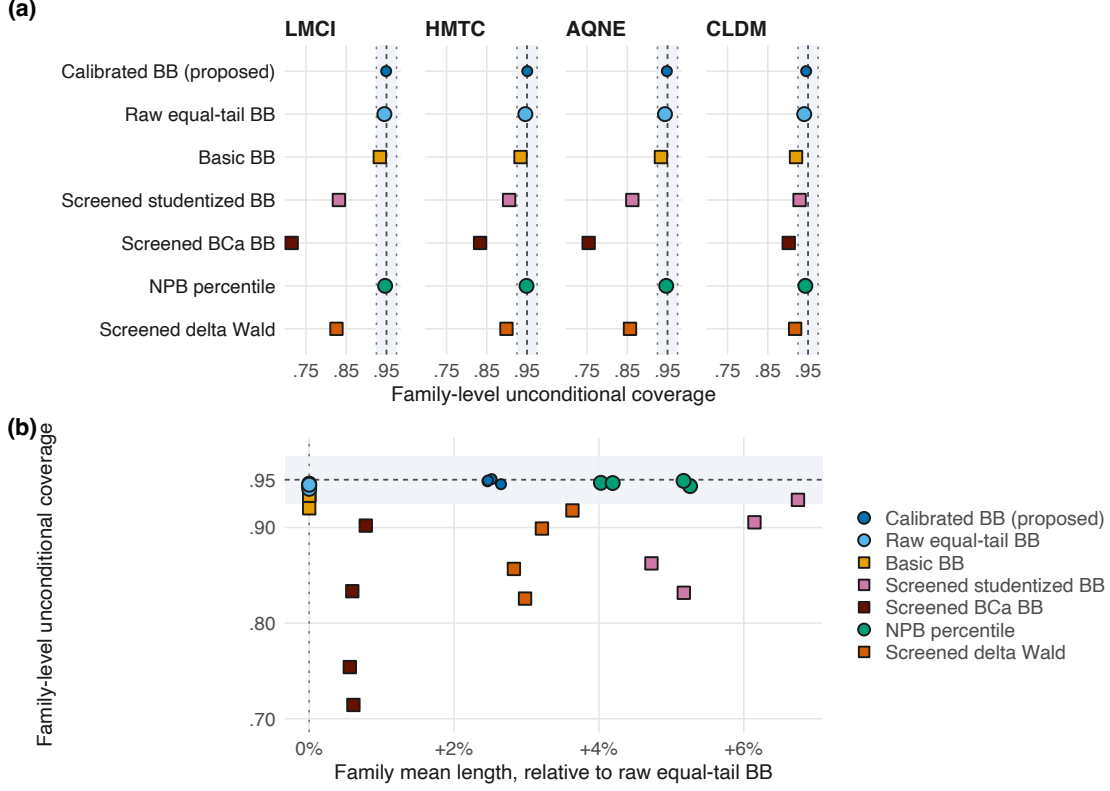
TABLE 1. Family-Level Unconditional Coverage and Mean Length in the Fresh Confirmatory Test

Interval method	Coverage (MCSE) ^a				Mean length ^c			
	LMCI	HMTC	AQNE	CLDM	LMCI	HMTC	AQNE	CLDM
Calibrated BB (proposed)	.9491 (.0014)	.9506 (.0014)	.9484 (.0014)	.9453 (.0018)	0.223	0.245	0.218	0.316
Raw equal-tail BB	.9447 (.0015)	.9459 (.0015)	.9437 (.0015)	.9404 (.0019)	0.218	0.239	0.213	0.308
Basic BB ^b	.9333 (.0016)	.9339 (.0016)	.9336 (.0016)	.9201 (.0021)	0.218	0.239	0.213	0.308
Screened studentized BB ^b	.8317 (.0022)	.9055 (.0018)	.8625 (.0021)	.9289 (.0020)	0.229	0.254	0.223	0.329
Screened BCa BB ^b	.7144 (.0022)	.8334 (.0021)	.7540 (.0022)	.9020 (.0023)	0.219	0.241	0.214	0.310
NPB percentile	.9465 (.0015)	.9488 (.0014)	.9467 (.0014)	.9433 (.0018)	0.227	0.252	0.222	0.324
Screened delta Wald ^b	.8257 (.0022)	.8989 (.0019)	.8567 (.0021)	.9179 (.0022)	0.224	0.247	0.219	0.319

Note. Fresh, untouched confirmatory test: 22 primary regular cells per interval method (4,000 scheduled replications each; 88,000 per method), evaluated with the frozen protocol. Coverage is unconditional over all scheduled replications; undefined intervals count as misses. The frozen family-level equivalence band is [.925, .975]. “Screened” methods are the strict-regularity-screened variants specified in the online supplement (Appendix C), whose typed refusals count as misses; they are not field-standard implementations. ^aFamily-level macro coverage with its Monte Carlo standard error in parentheses. ^bMethod failed at least one blocking same-gate criterion (coverage, catastrophic lower bound, or definedness); its length is reported for completeness but is not length-comparable to gate-passing methods. ^cEqual-cell conditional mean length across the family’s regular cells (defined intervals only). LMCI = latent monotone conditional independence; HMTC = heterogeneous-margin target control; AQNE = asymmetric-quality nonexchangeable items; CLDM = controlled local dependence. BB = Bayesian bootstrap; NPB = nonparametric bootstrap.

5.2.2. *Cell-Level Behavior of the Calibration.* Figure 4 disaggregates the raw-versus-calibrated comparison over all 22 regular cells. The correction behaves as Lemma 3.1 leads one to expect: the gain in coverage is concentrated where $\sqrt{n}$ is small (at $N = 60$ the mean cell coverage moves from .9335 (raw) to .9423 (calibrated)) and diminishes at large N , where both intervals are close to nominal and the $\gamma/\sqrt{n}$ widening has a negligible length cost. Across the 22 cells the calibrated interval’s coverage spans .939 to .959; the raw interval’s minimum is .931.

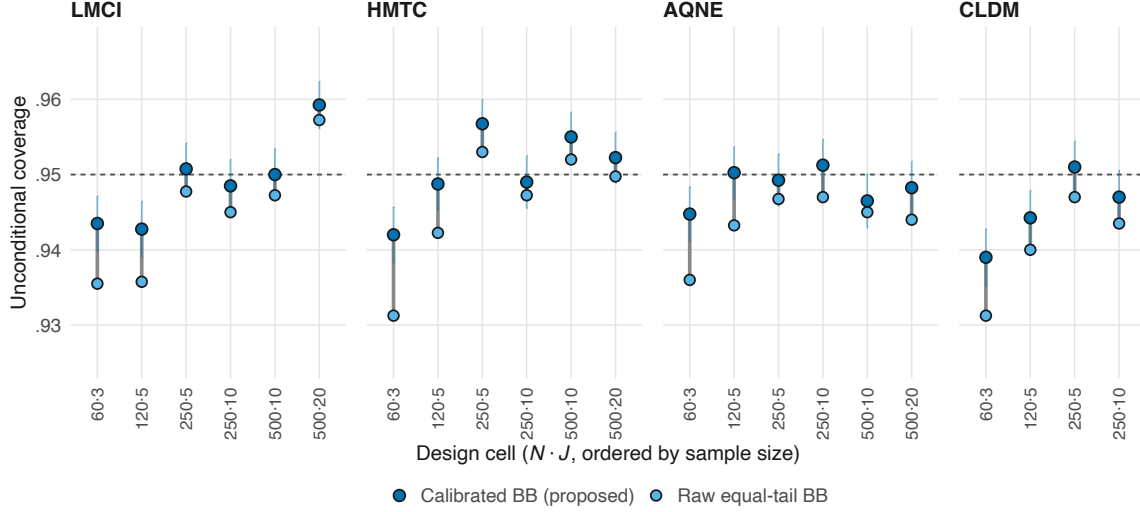
FIGURE 3. Family-Level Confirmatory Evidence for All Seven Interval Methods



Note. (a) Family-level unconditional coverage (undefined intervals count as misses) with ± 1 Monte Carlo standard error bars (mostly smaller than the plotting symbols), the nominal .95 line (dashed), and the frozen equivalence band [.925, .975] (shaded). Circles: methods that passed all blocking same-gate criteria; squares: methods that failed at least one. The calibrated interval (dark blue, enlarged) is the proposed method. (b) The trade-off between family mean length (relative to the raw equal-tail BB interval) and coverage; points right of 0% are longer than the raw interval. LMCI = latent monotone conditional independence; HMTc = heterogeneous-margin target control; AQNE = asymmetric-quality nonexchangeable items; CLDM = controlled local dependence; BB = Bayesian bootstrap; NPB = nonparametric bootstrap; BCa = bias-corrected and accelerated.

5.2.3. *Behavior of the Screened Comparators in the Regular Lane.* The comparator columns of Table 1 document a failure of definedness. The screened Wald interval's family coverage ranged from .826 to .918; the screened studentized BB behaved similarly (.832 to .929); and the screened BCa collapsed to family coverage as low as .714. The mechanism is visible at the cell level and, notably, is not confined to small samples: the most severe failures occur in large samples with many items. In the $N = 500$, $J = 20$, $H = .45$ cells, the margin ordering becomes crowded with

FIGURE 4. Raw Versus Calibrated Coverage in All 22 Primary Regular Cells



Note. Each column is one design cell, labeled $N \cdot J$ and ordered by sample size within family. Light circles: raw equal-tail BB coverage; dark circles: calibrated coverage; connecting segments show the lift. Error bars are ± 1 Monte Carlo standard error for the calibrated point ($\approx .0035$). Every displayed value lies inside the frozen equivalence band $[\cdot.925, \cdot.975]$; the dashed line marks nominal $\cdot.95$. Undefined intervals count as misses, and no cell produced any undefined BB interval (0 of 88,000 replications). Family abbreviations as in Figure 3.

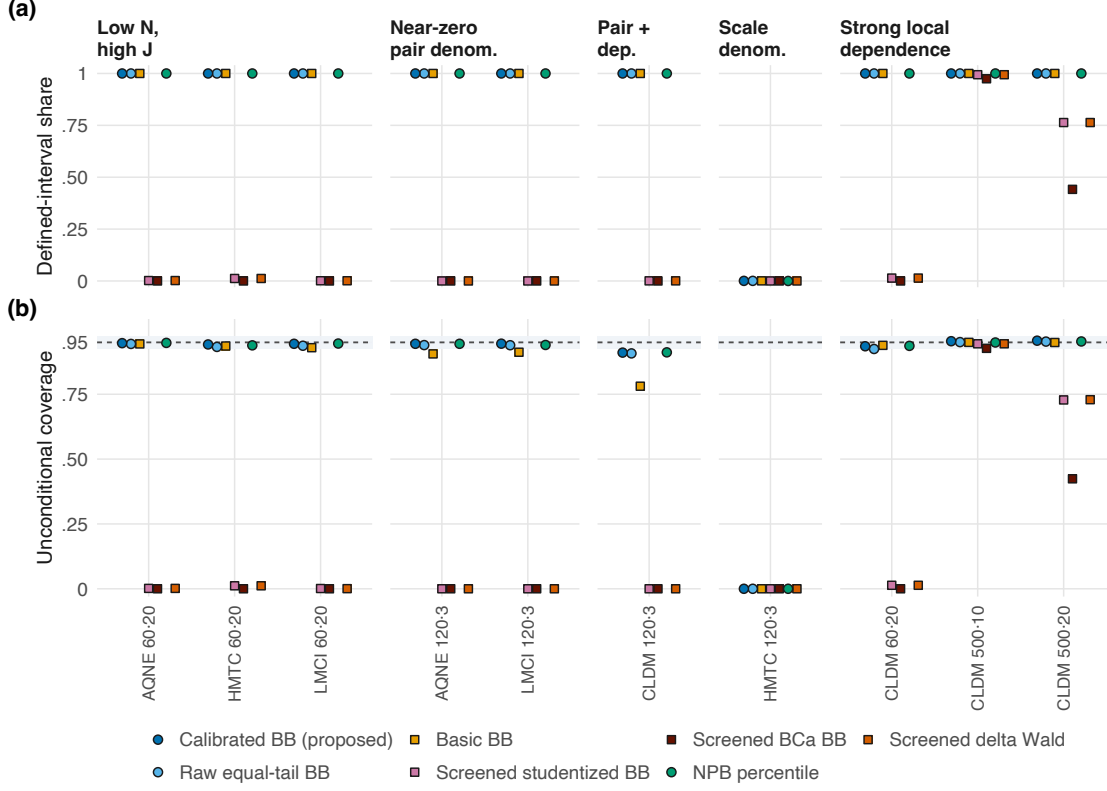
20 items, the min-kinks of Equation (2) lie close together, and every derivative- or jackknife-based construction encounters the screen: in the LMCI version of that cell the Wald interval covered $\cdot.469$, and BCa produced a defined interval in only 443 of 4,000 replications, for unconditional coverage of $\cdot.106$, in a cell that satisfies the frozen regularity thresholds. The failure is dominated by refused intervals rather than by misplaced ones: conditional on returning an interval, the screened BCa covered $\cdot.952$ to $\cdot.955$ and the screened Wald $\cdot.935$ to $\cdot.944$ at the family level. Near margin ties, then, derivative- and jackknife-based constructions must either decline to return an interval, as our screened variants do, or resolve the tie by convention, as released implementations do, and the finite-sample calibration of such conventions has not been established. The weighting- and resampling-based intervals face no such choice (calibrated: $\cdot.959$; raw: $\cdot.957$; NPB: $\cdot.958$ in the same cell), and across the entire regular lane the calibrated, raw, basic, and NPB intervals produced no undefined intervals at all, against 6,577 for the Wald and studentized constructions and 14,876 for BCa. In these conditions the derivative-free construction of the BB interval determines whether an interval exists at all.

5.2.4. *Stress Lane.* Figure 5 reports the 10 stress cells separately, as prespecified. Three patterns are apparent.

- (1) *Definedness collapses for the screened methods.* In the low- N , high- J cells ($N = 60$, $J = 20$) the SE- and jackknife-based methods lose definedness almost completely (the studentized interval was defined in as few as 2 of 2,000 replications), whereas the BB family and NPB remain defined and hold coverage near nominal.
- (2) *Local dependence does not break the interval.* Under strong local dependence (the CLDM stress variants) the BB family’s coverage remains between .93 and .95; dependence that violates the auxiliary measurement model does not, in these cells, invalidate the interval.
- (3) *The estimand has a boundary, and every method meets it.* In the near-zero scale-denominator cell, where $B(P) \approx 0$, the estimand itself approaches non-definability, intervals become undefined for all constructions, and unconditional coverage falls to zero. We report this behavior as it stands; it marks the point at which interval estimation of H ceases to be meaningful, and the typed failure semantics make that boundary visible to users.

5.2.5. *Monte Carlo Precision and Independent Validation.* Family-level coverage MC-SEs never exceeded .0022; the calibrated-raw coverage differences (+.0044 to +.0049) are paired contrasts on identical replications and draws and are therefore not attributable to Monte Carlo error. An independent validation pass (separate code path, recomputation of every summary in this section from the 648,000 stored endpoint rows, exact table comparison) returned 14 of 14 checks, and the gate evaluator confirmed 9 of 9 blocking criteria. Under the frozen protocol, the confirmatory prediction was borne out on untouched data. These results support calibration success within this frozen design; they are not a universal guarantee over all binary scales, item counts, sample sizes, or response regimes, and the stress lane indicates where the guarantee breaks down.

FIGURE 5. Stress-Lane Diagnostics: Definedness and Coverage at the Boundary



Note. The 10 prespecified stress cells, grouped by stress lane; within each lane, columns are cells labeled by family and $N \cdot J$. (a) Share of scheduled replications yielding a defined interval. (b) Unconditional coverage (undefined intervals count as misses), with the nominal .95 line and frozen band as in Figure 3. Circles: same-gate-eligible methods; squares: others. In the near-zero scale-denominator lane the estimand itself approaches non-definability and every method fails; full numbers in Appendix F. Family abbreviations as in Figure 3.

6. Empirical Illustration: The Scientific Reasoning Scale

6.1. Data and Provenance. The Scientific Reasoning Scale (SRS) of Drummond and Fischhoff (2017) measures the ability to evaluate scientific evidence with 11 true–false items. We analyze its Italian administration (Caliciuri & Lanz, 2024): an online sample of adults recruited in 2024, distributed publicly under a CC-BY 4.0 license. Of 624 received responses, 606 were consent-eligible and $n = 337$ provided complete responses to all 11 items; the analysis targets the complete-case response distribution of the fixed 11-item set.

The application was governed by the same outcome-blindness discipline as the simulation. The dataset and the fixed item set were selected, and the eligibility decision locked, before the source’s own psychometric report was read and before any coefficient or interval was computed, so that neither the choice of example nor the retention of all 11 items could be influenced by how favorable the result would appear. One consequence should be stated explicitly: the original authors ultimately recommended a reduced item subset for this population, and we do not use it, since a fixed, prespecified scale is the object our method addresses. Retaining the full set also makes the example a negative one.

Three working limitations attach to the data. Recruitment was nonprobability (snowball) sampling, so no population generalization is intended; the complete-case subset is the prespecified analysis frame, not a missing-data correction; and respondent independence is a working assumption, since no referral-network identifiers are available.

A further limitation concerns the transport of the simulation evidence, and the frozen contract requires us to state it: $(n, J) = (337, 11)$ lies inside the confirmatory design’s ranges (N 60–500, J 3–20) but is not itself a simulated cell, so the coverage evidence carries over to this configuration by interpolation rather than by direct match.

6.2. Results. The frozen algorithm ran without modification: 50,000 seeded BB draws, all defined, type-8 quantiles, $k_n = 1 + .25/\sqrt{337} = 1.0136$. [Figure 6](#) and [Table 2](#) summarize the outcome, and [Appendix G](#) reports the source verification, the analysis constants, the item-level descriptives, and the seed sensitivity. The fixed-scale point estimate is $\hat{H} = .1302$ (our ratio-of-sums implementation, the total-score identity used by the BB engine, and `mokken’s coefH` agree to 3×10^{-15}), with raw 95% interval $[.1029, .1614]$ and calibrated interval $[.1025, .1618]$. At $n = 337$ the calibration is modest (+1.36% width), as [Lemma 3.1](#) implies: the correction matters most at small n and nearly vanishes here.

The substantive interpretation is clear. The entire calibrated interval lies below the conventional .30 reference, a comparison that we report descriptively rather than as a formal threshold decision or scale classification, and the interval indicates that the data are sufficient to make this placement stable rather than an artifact of point-estimate variability. The item-level and pair-level descriptive statistics in [Figure 6a–b](#) indicate why; no interval is computed for them, no selection is performed, and the fixed scale is not searched. Item coefficients range from .053 to .229, five of 55 pair

coefficients are negative, and the strong SRS8–SRS10 pair ($H_{ij} = .64$) stands alone in an otherwise weakly associated matrix. The result is also computationally stable: across three independent 50,000-draw seeds the calibrated endpoints moved by at most .0007, endpoint Monte Carlo standard errors are below .0002, and reversing the item order reproduces the interval to numerical precision (Table 2).

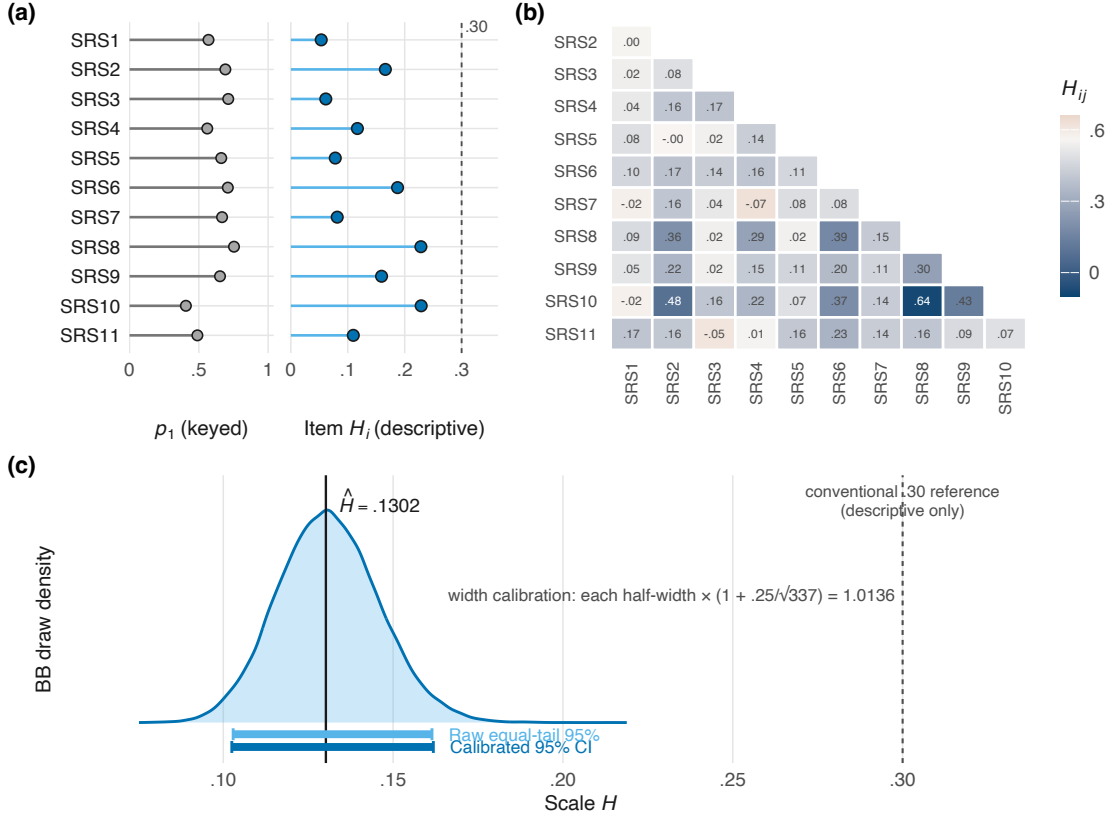
An interval method intended for applied reporting must be able to return an unfavorable answer, and this application demonstrates that behavior: no retuning was performed, the example was not selected for its outcome, and a clear negative finding is reported with defensible precision. For comparison, **mokken**’s analytic standard error here is .0148, and a Wald interval built from it is similar in width to ours at this n ; the two constructions are distinguished by the simulation evidence rather than by this single example.

TABLE 2. Scale-Level Results for the Fixed 11-Item Scientific Reasoning Scale ($n = 337$)

Quantity	Value
Point estimate $\hat{H}$	.1302
mokken analytic SE	.0148
Raw equal-tail 95% BB interval	[.1029, .1614]
Calibrated 95% interval	[.1025, .1618]
Width multiplier $k_n = 1 + .25/\sqrt{337}$	1.0136
Interval length (raw $\rightarrow$ calibrated)	.0585 $\rightarrow$.0593
Calibrated endpoint MCSE (lower; upper)	.00017; .00019
<i>Calibrated interval under independent seeds (50,000 draws each)</i>	
Seed 2026072611	[.1025, .1618]
Seed 2026072612	[.1025, .1611]
Seed 2026072613	[.1028, .1617]
<i>Calibrated interval under draw-count prefixes (seed 2026072611)</i>	
5,000 draws	[.1026, .1615]
20,000 draws	[.1026, .1615]
50,000 draws	[.1025, .1618]

Note. All quantities computed on the prespecified complete-case 11-item scale with the frozen algorithm (respondent Dirichlet weights, type-8 equal-tail quantiles, $\gamma = .25$, no endpoint clipping). All 50,000 draws were defined. The analytic standard error is the delta-method value reported by **mokken** for reference. Both intervals lie entirely below the conventional .30 reference; that comparison is descriptive, not a formal threshold decision.

FIGURE 6. The Fixed 11-Item Scientific Reasoning Scale ($n = 337$)



Note. (a) Descriptive item statistics for the fixed item set: proportion answering in the keyed direction (p_1) and item scalability coefficients H_i ; the dashed line marks the conventional .30 reference. No item-level interval is computed and no item is selected or removed. (b) All 55 descriptive pair coefficients H_{ij} ; five pairs are negative (warm cells). (c) The BB distribution of scale H from 50,000 draws, the sample estimate ($\hat{H} = .1302$, solid line), the raw equal-tail and calibrated 95% intervals, and the descriptive .30 reference. The calibrated interval lies entirely below .30. BB = Bayesian bootstrap; CI = confidence interval.

7. Discussion

7.1. Summary of Findings. For the fixed binary scale-level Mokken coefficient, this article supplies an interval with demonstrated finite-sample coverage, obtained by the smallest defensible modification of a first-order-justified procedure. The division of labor among the components is deliberate. The Bayesian bootstrap contributes a derivative-free engine whose conditional law provably matches first-order sampling theory (Proposition 3.2). The width calibration contributes a deterministic $O_p(n^{-1})$ endpoint perturbation that preserves the first-order limit (Corollary 3.1), can never reduce coverage (Proposition 3.3), and costs exactly $\gamma/\sqrt{n}$ in relative length. The

tuning constant $\gamma = .25$ contributes the single empirical degree of freedom, fixed on training data, selected on held-out data, locked, and then confirmed prospectively, with family-level coverage of 94.5% to 95.1% in a fresh four-family design whose acceptance criteria were recorded in advance. The empirical application shows the frozen method returning a stable and substantively unfavorable answer without modification.

The confirmatory study yielded a second finding that we had not anticipated when the design was frozen. Our screened influence-function Wald and BCa comparators undercovered in every family, and the failures were not confined to small samples: with many items, near-tied margins interact with the coefficient’s min-kinks in a way that affects every construction that differentiates or jackknives, and in a regular $N=500$, $J=20$ cell the screened Wald interval covered less than half its nominal level while the screened BCa interval was almost never defined. The failure consisted almost entirely of refusals rather than misplaced intervals. Released implementations resolve margin ties by convention instead of refusing (for example, `mokken` averages the admissible tied orderings into a finite standard error), and whether such conventions are well calibrated near the kinks is a question our results raise; applied users with long scales are the ones most affected by it. The practical conclusion we draw is that, for fixed binary scales, $\hat{H}$ should be reported with a reweighting-based interval (the calibrated interval proposed here, or at minimum the NPB percentile interval) whose definedness does not depend on a tie-resolution convention, and that an undefined interval should be treated as information about the data rather than as a nuisance.

7.2. Practical Guidance. Several recommendations for reporting practice follow. First, the calibrated interval should be presented alongside $\hat{H}$ whenever a scalability claim matters, particularly when the estimate lies near the conventional bands; the SRS application illustrates how an interval converts a fragile point comparison into a stable statement. Second, the conventional .30, .40, and .50 references are best treated as descriptive landmarks: a decision-grade procedure for claims of the form $H \geq .30$ requires its own error guarantees, is deliberately not improvised here, and is the natural next step discussed below. Third, the interval presumes a scale that was fixed in advance. In the common workflow in which items are first selected by the automated search procedure and H is then reported for the selected set, the reported coefficient raises a selection-inference problem that no interval in this article addresses. Fourth, the failure semantics should be respected: the method declines to produce an interval when the weighted denominator degenerates, and the stress lane indicates

that this occurs precisely where no method has a defensible answer, since near-zero scale denominators mark the boundary of the estimand itself. The `bayesmokken` call in the Software subsection provides a direct adoption path; at $B = 2,000$ draws the computation is interactive even at $J = 20$, $N = 500$.

7.3. Scope and Limitations. The interval is a BB-based confidence interval: we use the Dirichlet weighting as a repeated-sampling approximation device, make no posterior-probability statements, and none of our validation would license them. The constant $\gamma = .25$ is simulation-calibrated, not analytically derived; the curvature identity (Equation (6)) fixes the order of the correction but neither its form, its direction, nor its value, and we prove no second-order coverage-accuracy theorem. The confirmatory result is what it states: coverage inside a frozen equivalence band across four prespecified families and 22 regular cells. This is evidence of transportable behavior rather than a universal theorem, and the stress lane shows where it stops. Finally, the scope is exactly the estimand of Equation (3): a fixed, prespecified binary item set, complete responses, independent respondents, on the regular domain. Exact marginal ties, near-zero denominators, item selection, missing data, ordinal responses, and clustered or otherwise dependent respondents all lie outside the present evidence.

7.4. Future Directions. Several extensions appear worth pursuing. First, second-order theory: an Edgeworth or von Mises expansion of the raw BB interval’s coverage error for this quotient functional would either explain the empirical adequacy of a constant γ or refine it into an analytic, possibly P -dependent, choice. Second, the threshold branch: applied users ultimately want calibrated evidence about claims of the form $H \geq .30$, and a decision-grade version of that machinery, with its own error guarantees distinct from the interval reported here, is a natural sequel. Third, the nonregular boundary: directional-derivative asymptotics (Fang & Santos, 2019) offer a route to valid inference at marginal ties, where the present theory intentionally stops. Fourth, richer designs: polytomous items and clustered respondents each change the estimand, the weight law, and the failure geometry, and in our view they require, and are receiving in dedicated companion work, their own frozen estimands, hierarchical weight laws, and validation chains rather than an informal transfer of the binary result. The two-level scalability literature (Koopman, Zijlstra, & van der Ark, 2020; Koopman et al., 2022) marks the analytic frontier that such an extension must meet.

7.5. Concluding Remarks. A methodological point deserves emphasis. Everything reported here (constants, gates, families, budgets, and seeds) was locked before the

data that judged it existed, in hash-verified protocol files that ship with the replication archive, consistent with a strict reading of the transparency literature (Depaoli & van de Schoot, 2017; Nosek et al., 2015; Wilson et al., 2014). Methods papers routinely ask applied researchers to preregister, and the discipline benefits when its methodologists accept the same constraint. The coverage figures in this article are predictions that were subsequently evaluated on untouched data, and we hope that this practice, as much as any single interval, proves useful: uncertainty statements for measurement coefficients can themselves be constructed and audited in the manner of confirmatory science.

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Software and reproducibility. The analyses used R 4.6.0 (R Core Team, 2026); the calibrated interval is implemented in the R package `bayesmokken` (Lee, 2026), and point-estimate concordance was checked against `mokken` (van der Ark, 2007). Every stage of the study (candidate grids, selection rules, acceptance gates, data-generating parameters, seeds, and evaluation code) was serialized into hash-locked protocol and receipt files before the outcomes it governs were generated; those files, their SHA-256 digests, the 648,000 stored endpoint rows, and the scripts that verify them ship with the replication archive. The package is available at <https://github.com/joonho112/bayesmokken> and the replication archive at <https://github.com/joonho112/bayesmokken-scalability-replication>.

Data availability. The empirical data are public: the Italian administration of the Scientific Reasoning Scale (Caliciuri & Lanz, 2024) is distributed under a CC-BY 4.0 license through OSF project jk9dp, <https://osf.io/jk9dp/>. The analysis consumed a checksummed local mirror of the released files without modification.

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Online Supplement

A Calibrated Bayesian Bootstrap Confidence Interval for Mokken’s Scale-Level Scalability Coefficient

JoonHo Lee

This supplement accompanies the main text. Section, equation, table, and figure numbers below are prefixed by the appendix letter (A.1, B.1, ...). References to the main text use its unprefix numbering.

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Appendix A. Notation, Estimand, and Equivalent Forms

A.1. Setting and Moments. A fixed, prespecified set S of J binary items produces response vectors $X = (X_1, \dots, X_J) \in \{0, 1\}^J$ with unknown population law P on the finite support $\{0, 1\}^J$. For $i \neq j$,

$$p_i(P) = \mathbb{E}_P X_i, \quad m_{ij}(P) = \mathbb{E}_P (X_i X_j). \quad (\text{A.1})$$

Write $\mathcal{E}_S = \{(i, j) : i < j\}$ with $K = \binom{J}{2}$ pairs. The pair numerator and denominator, aggregates, and estimand are

$$C_{ij}(P) = m_{ij}(P) - p_i(P)p_j(P), \quad D_{ij}(P) = \min\{p_i(P), p_j(P)\} - p_i(P)p_j(P), \quad (\text{A.2})$$

$$A(P) = \sum_{(i,j) \in \mathcal{E}_S} C_{ij}(P), \quad B(P) = \sum_{(i,j) \in \mathcal{E}_S} D_{ij}(P), \quad H(P) = \frac{A(P)}{B(P)} \quad \{B(P) > 0\}. \quad (\text{A.3})$$

The estimator is the plug-in $\hat{H} = H(P_n)$ with $P_n = n^{-1} \sum_{r=1}^n \delta_{X_r}$; empirical moments use the $1/n$ convention (an $n - 1$ covariance divisor is *not* the defining numerator). D_{ij} is the maximum covariance compatible with the two margins, the comonotone (common-quantile) upper bound of Rüschemdorf (1982) (see also Puccetti and Wang, 2015), so H is observed association relative to its margin-permitted ceiling.

A.2. The Scale Coefficient Is a Weighted Ratio of Sums. When all pair denominators are positive,

$$H(P) = \sum_{(i,j) \in \mathcal{E}_S} w_{ij}(P) H_{ij}(P), \quad w_{ij}(P) = \frac{D_{ij}(P)}{B(P)}, \quad H_{ij} = \frac{C_{ij}}{D_{ij}}, \quad (\text{A.4})$$

a denominator-weighted combination that in general differs from the unweighted pair average $K^{-1} \sum H_{ij}$. All derivatives, influence functions, and interval procedures in this project address the weighted ratio-of-sums functional, never the pair average.

A.3. Equivalence With the Guttman-Error Form. Order the items within each pair so that $p_\ell \leq p_u$ (the harder item ℓ has the smaller success probability; on the regular domain the ordering is strict and unique). Define the pairwise Guttman-error probability and its independence benchmark,

$$f_{\ell u} = \mathbb{E}\{X_\ell(1 - X_u)\} = p_\ell - m_{\ell u}, \quad e_{\ell u}^0 = p_\ell(1 - p_u). \quad (\text{A.5})$$

Then $C_{ij} = m_{\ell u} - p_\ell p_u = e_{\ell u}^0 - f_{\ell u}$ and $D_{ij} = p_\ell - p_\ell p_u = e_{\ell u}^0$, so summing over pairs,

$$H = \frac{\sum (e_{\ell u}^0 - f_{\ell u})}{\sum e_{\ell u}^0} = 1 - \frac{F}{E_0}, \quad F = \sum_{\text{pairs}} f_{\ell u}, \quad E_0 = \sum_{\text{pairs}} e_{\ell u}^0, \quad (\text{A.6})$$

the classical “one minus observed over expected Guttman errors” form (Loevinger, 1948; Mokken, 1971). The two representations are algebraically identical on the regular domain; the ratio-of-covariances form Equation (A.3) is the one we differentiate.

A.4. Domains. Let $\mathcal{P}_J$ denote the probability simplex on $\{0, 1\}^J$. The functional domain is $\mathcal{D}_H = \{P : B(P) > 0\}$. For a permutation π of item labels, the open order cell is

$$\mathcal{R}_\pi = \{P \in \mathcal{P}_J : 0 < p_{\pi(1)}(P) < \cdots < p_{\pi(J)}(P) < 1\}, \quad (\text{A.7})$$

on which every pair’s active-min item is unique and locally constant and $D_{ij}(P) = p_{\ell_{ij}}(P)\{1 - p_{u_{ij}}(P)\} > 0$. The ordinary regular domain is $\mathcal{R} = \bigcup_\pi \mathcal{R}_\pi$; uniform statements use separated subsets $\mathcal{R}_\pi(\varepsilon, \delta, b)$ requiring margins in $[\varepsilon, 1 - \varepsilon]$, consecutive margin gaps at least δ , and $B \geq b$. Full response-pattern support is not required.

Marginal ties, boundary margins, and $B = 0$ are outside $\mathcal{R}$; the evaluation protocol types these states explicitly rather than repairing them (Appendix C).

Appendix B. Proofs and Curvature Identities

Throughout, h denotes a signed measure on $\{0, 1\}^J$ with $\sum_x h_x = 0$ (a tangent direction), and $\dot{p}_i[h] = \sum_x x_i h_x$, $\dot{m}_{ij}[h] = \sum_x x_i x_j h_x$ are the induced moment derivatives. All statements are local to a fixed order cell $\mathcal{R}_\pi$ (Appendix A).

B.1. Derivatives and the Influence Function. For any P and direction h ,

$$\dot{C}_{ij,P}[h] = \dot{m}_{ij}[h] - p_j(P) \dot{p}_i[h] - p_i(P) \dot{p}_j[h], \quad (\text{B.1})$$

and inside $\mathcal{R}_\pi$, with $\ell = \ell_{ij}$ and $u = u_{ij}$ the active lower- and upper-margin items of the pair,

$$\dot{D}_{ij,P}[h] = \{1 - p_u(P)\} \dot{p}_\ell[h] - p_\ell(P) \dot{p}_u[h]. \quad (\text{B.2})$$

Both maps are linear in h . Summing over pairs gives $\dot{A}_P[h]$ and $\dot{B}_P[h]$, and the quotient rule gives

$$\dot{H}_P[h] = \frac{\dot{A}_P[h] - H(P) \dot{B}_P[h]}{B(P)}. \quad (\text{B.3})$$

Evaluating at the one-point direction $h = \delta_y - P$ yields the influence functions

$$\begin{aligned} \phi_A(y; P) &= \dot{A}_P[\delta_y - P], & \phi_B(y; P) &= \dot{B}_P[\delta_y - P], \\ \psi_H(y; P) &= \frac{\phi_A(y; P) - H(P) \phi_B(y; P)}{B(P)}. \end{aligned} \quad (\text{B.4})$$

Explicitly, writing $y = (y_1, \dots, y_J)$, $\phi_{C,ij}(y; P) = (y_i y_j - m_{ij}) - p_j(y_i - p_i) - p_i(y_j - p_j)$ and $\phi_{D,ij}(y; P) = (1 - p_u)(y_\ell - p_\ell) - p_\ell(y_u - p_u)$, summed over pairs. Centering $\mathbb{E}_P \psi_H(Y; P) = 0$ is immediate because each $\dot{p}_i[\delta_Y - P]$ and $\dot{m}_{ij}[\delta_Y - P]$ has mean zero. These formulas were verified against central-difference numerical oracles at machine precision (1,561 checks, zero failures; maximum absolute derivative error $< 1.6 \times 10^{-8}$; maximum influence-centering error $< 3.9 \times 10^{-16}$).

B.2. Asymptotic Linearity. The following restates Proposition 3.1 of the main text.

Proposition B.1. *If $P \in \mathcal{R}_\pi$ with $B(P) > 0$ and $\sigma_H^2 = \text{Var}_P\{\psi_H(Y; P)\} \in (0, \infty)$, then $H(P_n) - H(P) = n^{-1} \sum_r \psi_H(X_r; P) + o_p(n^{-1/2})$ and $\sqrt{n}\{H(P_n) - H(P)\} \xrightarrow{d} \mathcal{N}(0, \sigma_H^2)$.*

Proof. The moment vector $T(P) = (p_1, \dots, p_J, m_{12}, \dots, m_{J-1,J})$ lives in a finite-dimensional Euclidean space, $T(P_n)$ is a sample mean of bounded vectors, and $\sqrt{n}\{T(P_n) - T(P)\}$ obeys the multivariate CLT. On $\mathcal{R}_\pi$ the map $T \mapsto H$ is a composition of linear maps, products, the coordinate selection $\min\{p_i, p_j\} = p_{\ell_{ij}}$ (locally constant selection, hence locally linear), and a quotient with denominator bounded away from zero on a neighborhood of $T(P)$; it is therefore continuously differentiable there, with gradient assembled from Equation (B.1)–Equation (B.3). The delta method delivers both claims, and the gradient’s inner product with the centered moment vector of a single observation is exactly $\psi_H(y; P)$ of Equation (B.4). $\square$

B.3. First-Order BB Matching. The following restates Proposition 3.2 of the main text.

Proposition B.2. *Under the conditions of Proposition B.1, with $(W_1, \dots, W_n) \sim \text{Dirichlet}(1, \dots, 1)$ independent of the data and $P_n^W = \sum_r W_r \delta_{X_r}$, the conditional law of $\sqrt{n}\{H(P_n^W) - H(P_n)\}$ given $X_{1:n}$ converges in probability to $\mathcal{N}(0, \sigma_H^2)$.*

Proof sketch. Write $G_r = W_r - 1/n$. The BB moment perturbation is $T(P_n^W) - T(P_n) = \sum_r G_r t(X_r)$ with $t(x)$ the bounded moment coordinates of a single observation. Flat Dirichlet weights satisfy $\mathbb{E}W_r = 1/n$, $\text{Var}(W_r) = (n-1)/\{n^2(n+1)\}$, and $\text{Cov}(W_r, W_s) = -1/\{n^2(n+1)\}$ for $r \neq s$; the conditional multivariate CLT for Dirichlet-weighted means (Lo, 1987) then gives, in probability, $\sqrt{n}\{T(P_n^W) - T(P_n)\} \mid X_{1:n} \rightsquigarrow \mathcal{N}(0, \Sigma_T(P))$ with the same covariance Σ_T as the sampling CLT for $\sqrt{n}\{T(P_n) - T(P)\}$, because the empirical covariance of $t(X_r)$ converges to Σ_T almost surely. Since $T(P_n) \rightarrow T(P)$ almost surely and H is continuously differentiable at $T(P)$ with the gradient of Proposition B.1, the conditional delta method applied along almost every sample path transfers the limit to H , with asymptotic variance $\nabla H^\top \Sigma_T \nabla H = \sigma_H^2$. The archived derivation (replication package, theory/) spells out the uniform-integrability and path-regularity details. $\square$

B.4. Quotient Curvature and the Exact Dirichlet Contraction. Within a regular cell each C_{ij} is bilinear in moments and each D_{ij} is a product of two margin coordinates, so second directional derivatives exist and, along zero-mass directions Q, R ,

$$A_P''[Q, R] = B_P''[Q, R] = - \sum_{i < j} (\dot{p}_i[Q] \dot{p}_j[R] + \dot{p}_i[R] \dot{p}_j[Q]) =: \mathcal{Q}_P[Q, R], \quad (\text{B.5})$$

(for A : the second derivative of m_{ij} vanishes and only the $-p_i p_j$ term curves; for $D_{ij} = p_\ell(1 - p_u)$: the bilinear cross term, with the locally constant active-min selection). The quotient Hessian follows from twice differentiating A/B :

$$H_P''[Q, R] = \frac{A_P''[Q, R]}{B} - \frac{A_P'[Q] B_P'[R] + A_P'[R] B_P'[Q] + A B_P''[Q, R]}{B^2} + \frac{2A B_P'[Q] B_P'[R]}{B^3}. \quad (\text{B.6})$$

Proposition B.3 (Exact local contraction). *Let $z_r = \delta_{X_r} - P_n$ and $\hat{\kappa}_r = H_{P_n}''[z_r, z_r]$. Then*

$$\mathbb{E}_W \left\{ \frac{1}{2} H_{P_n}''[P_n^W - P_n, P_n^W - P_n] \right\} = \frac{1}{2n(n+1)} \sum_{r=1}^n \hat{\kappa}_r. \quad (\text{B.7})$$

Proof. Because $\sum_r (W_r - 1/n) = 0$, we may write $P_n^W - P_n = \sum_r (W_r - 1/n) z_r$. Bilinearity gives

$$\mathbb{E}_W \left\{ \frac{1}{2} H''[P_n^W - P_n, P_n^W - P_n] \right\} = \frac{1}{2} \sum_{r,s} \mathbb{E} \{ (W_r - \frac{1}{n})(W_s - \frac{1}{n}) \} H''[z_r, z_s].$$

Substituting $\text{Var}(W_r) = (n-1)/\{n^2(n+1)\}$ and $\text{Cov}(W_r, W_s) = -1/\{n^2(n+1)\}$,

$$= \frac{1}{2n^2(n+1)} \left\{ n \sum_r H''[z_r, z_r] - H'' \left[\sum_r z_r, \sum_s z_s \right] \right\} = \frac{1}{2n(n+1)} \sum_r \hat{\kappa}_r,$$

since $\sum_r z_r = 0$ annihilates the second term. $\square$

The identity was verified numerically to 2.4×10^{-17} . Its role in the article is strictly motivational: it fixes the *order* (n^{-1}) of a finite-sample correction and nothing else. The midpoint-symmetric width map is a design candidate adopted at that order; neither the shape, nor the direction, nor the value of γ follows from the identity.

B.5. Exact Properties of the Width Map. The three statements below restate [Proposition 3.3](#), [Lemma 3.1](#), and [Corollary 3.1](#) of the main text.

Proposition B.4. *For $n \geq 1$, $\gamma \geq 0$, and any defined interval $L_n \leq U_n$ with $m_n = (L_n + U_n)/2$, $w_n = (U_n - L_n)/2$, and $[\tilde{L}_n, \tilde{U}_n] = [m_n - k_n w_n, m_n + k_n w_n]$, $k_n = 1 + \gamma/\sqrt{n}$: (a) the midpoint is m_n ; (b) $[L_n, U_n] \subseteq [\tilde{L}_n, \tilde{U}_n]$; (c) $(\tilde{U}_n - \tilde{L}_n)/(U_n - L_n) = k_n$ when $U_n > L_n$; (d) $\mathbf{1}\{\theta \in [L_n, U_n]\} \leq \mathbf{1}\{\theta \in [\tilde{L}_n, \tilde{U}_n]\}$ for every θ .*

Proof. $k_n \geq 1$ and $w_n \geq 0$ give $m_n - k_n w_n \leq m_n - w_n = L_n$ and $U_n = m_n + w_n \leq m_n + k_n w_n$, which is (a), (b), and (d); (c) is $2k_n w_n / 2w_n$. $\square$

Lemma B.1. *If γ is fixed and $w_n = O_p(n^{-1/2})$, then $\tilde{L}_n - L_n = -\gamma w_n / \sqrt{n} = O_p(n^{-1})$ and $\tilde{U}_n - U_n = \gamma w_n / \sqrt{n} = O_p(n^{-1})$; hence $\sqrt{n}(\tilde{L}_n - L_n) = o_p(1)$ and $\sqrt{n}(\tilde{U}_n - U_n) = o_p(1)$.*

Proof. The endpoint differences are exactly $\mp \gamma w_n / \sqrt{n}$ by definition; multiply by $\sqrt{n}$. $\square$

Corollary B.1. *If in addition $(\sqrt{n}\{L_n - \theta\}, \sqrt{n}\{U_n - \theta\}) \rightsquigarrow (Z_L, Z_U)$ jointly with $\mathbb{P}(Z_L = 0 \text{ or } Z_U = 0) = 0$, then the calibrated endpoint vector has the same limit and $\mathbb{P}\{L_n \leq \theta \leq U_n\} - \mathbb{P}\{\tilde{L}_n \leq \theta \leq \tilde{U}_n\} \rightarrow 0$.*

Proof. By [Lemma B.1](#) and Slutsky, the calibrated endpoint vector converges to the same (Z_L, Z_U) . Coverage is $\mathbb{P}(Z_L^{(n)} \leq 0 \leq Z_U^{(n)})$ in the local parameterization; the no-mass-on-boundary condition makes the coverage functional continuous at the limit law, so both coverages converge to $\mathbb{P}(Z_L \leq 0 \leq Z_U)$. $\square$

Remark B.1. [Proposition B.4\(d\)](#) is a per-replication statement and therefore holds under *any* data-generating process, including every stress cell: whenever the raw interval is defined and covers, the calibrated interval covers. Combined with the inherited failure semantics (an undefined raw interval stays undefined), unconditional coverage of the calibrated interval can never fall below that of the raw interval on the same replications. None of these results claims second-order accuracy, an Edgeworth-error cancellation, or any optimality of $\gamma = .25$.

Appendix C. Algorithm and Comparator Specifications

C.1. The Locked Primary Method. The public method identifier is P7-FS-WIDTH-RAW-G025-v1. Given the $n \times J$ complete binary matrix, level .95, and B draws:

- (1) Draw B weight vectors $(W_1, \dots, W_n) \sim \text{Dirichlet}(1, \dots, 1)$ from a seeded, draw-major stream (Gamma(1,1) variates normalized by their sum; the stream is reproducible across batch sizes and worker counts).
- (2) For each draw, compute weighted margins $p_i^W = \sum_r W_r X_{ri}$ and joint moments $m_{ij}^W = \sum_r W_r X_{ri} X_{rj}$, then $H(P_n^W) = A(P_n^W)/B(P_n^W)$ by the ratio-of-sums formula. A draw with $B(P_n^W) \leq 0$ is typed UNDEFINED_DENOMINATOR.
- (3) If the number of defined draws meets the frozen minimum, the raw interval $[L_n, U_n]$ is the type-8 (median-unbiased) quantile pair at (.025, .975) of the defined draws; otherwise the replication is typed undefined.

- (4) The reported interval is $[m_n - k_n w_n, m_n + k_n w_n]$ with $k_n = 1 + .25/\sqrt{n}$. No endpoint clipping; the base method’s typed status is inherited; a per-replication cascade to any other method is forbidden.

In every performance summary, coverage denominators are all scheduled replications, so that an undefined interval counts as a miss. Conditional mean lengths (defined draws only) are labeled as such wherever reported.

C.2. Comparator Registry. All comparators run on the same replications with the same failure accounting; the BB-based comparators share the same $B = 2,000$ weight draws as the primary method’s base, and the NPB uses $B = 2,000$ index-addressed respondent resamples. [Table C.1](#) transcribes the frozen registry entries.

TABLE C.1. Frozen Comparator Specifications

Method (ID)	Exact algorithm
Raw equal-tail BB (CMP-RAW-BB-EQUAL- TAIL-v1)	Type-8 quantiles [.025, .975] of $B=2,000$ common respondent Dirichlet(1) H draws.
Basic BB (CMP-BASIC-BB-v1)	$2\hat{H}$ minus the reversed type-8 quantiles [.975, .025] of the same BB draws (reflection interval).
Studentized BB (CMP-ANALYTIC- STUDENTIZED-BB-v1)	Reversed type-8 quantiles of $(H_b - \hat{H})/\widehat{SE}_b$, where $\widehat{SE}_b$ is the analytic influence-function standard error evaluated on each reweighted distribution; pivots re-anchored at $\hat{H} \pm q \widehat{SE}$. Undefined whenever the per-draw studentizer is undefined or degenerate.
BCa BB (CMP-BCA-BB-v1)	Standard BCa probability transform using the BB draws for bias correction (z_0) and respondent delete-one jackknife acceleration; type-8 quantiles at the adjusted probabilities. Undefined when z_0 or the acceleration is undefined or the adjusted probabilities degenerate.
NPB percentile (CMP-NPB-PERCENTILE-v1)	Type-8 quantiles [.025, .975] of $B=2,000$ index-addressed respondent resamples with replacement.
Delta-method Wald (CMP-DELTA-IF-WALD-v1)	$\hat{H} \pm z_{.975} \cdot \widehat{SE}_{\text{IF}}$ with the analytic influence-function standard error of Equation (B.4) ; no clipping. Undefined when the standard error is undefined (e.g., at an exact empirical margin tie or empty active-min branch).

Two optional secondary constructions in the registry (NPB basic and standard NPB BCa) were not part of the confirmatory efficiency lane and are not reported.

The delta-method Wald construction is built in the spirit of analytic-SE practice (Kuijpers et al., 2013; van der Ark, 2007), and the bootstrap constructions follow DiCiccio and Efron (1996). However, all three derivative- or jackknife-based comparators here are *strict-regularity-screened variants*: at an exact empirical margin tie, an empty active-min branch, or a degenerate transform they return a typed undefined interval rather than resolving the tie by convention. Released implementations make the opposite choice (mokken’s analytic standard error, for instance, averages the admissible tied orderings and returns a finite value), so the comparator results in this article characterize the screened variants specified above, not the behavior of any released package; no field implementation was evaluated.

C.3. Typed Failure Taxonomy. Every replication carries one typed status per method. The taxonomy distinguishes, at minimum: success; an undefined weighted or resampled scale denominator ($B \leq 0$ on the perturbed distribution); an insufficient number of defined draws; a degenerate or undefined studentizer (studentized interval only); a degenerate bias-correction or acceleration transform (BCa only); an undefined analytic standard error (Wald only, e.g., at an exact empirical margin tie); and nonfinite or reversed endpoints. Statuses are recorded verbatim in the raw endpoint files and audited by the independent validator; no status is ever silently converted to a success, and the exact status vocabulary is part of the hash-locked protocol in the replication archive.

Appendix D. Data-Generating Families and Truth Oracles

D.1. Mechanisms. All four families generate independent respondents with complete responses on a fixed item set; population truth is computed analytically or by audited quadrature, never by simulation.

LMCI (latent monotone conditional independence). $\theta \sim \mathcal{N}(0, 1)$ and $\mathbb{P}(Y_j = 1 \mid \theta) = \text{logit}^{-1}(b_j + s a_j \theta)$ with $a_j > 0$: monotone, conditionally independent response functions, so that the data satisfy the monotone homogeneity model. Marginal and pairwise moments come from adaptive normal quadrature; the target H is matched by a bracketed scalar root in the signal strength s , with the bracket expansion audited. HMTC (heterogeneous-margin target control). Two latent classes with mixing weight π and $\mathbb{P}(Y_j = 1 \mid L) = q_j + \lambda d_j (L - \pi)$, so each item margin equals q_j *exactly* while pair covariances are $\lambda^2 \pi (1 - \pi) d_i d_j$. The ratio-of-pair-sums identity then yields the target H in closed form. Margin configurations, the raw material of the min denominators,

are therefore controlled directly; the near-zero scale-denominator stress cell places all margins within 10^{-6} to 5×10^{-6} of one another with a rare latent class.

AQNE (asymmetric-quality nonexchangeable items). Finite latent classes at locations z_c with $\mathbb{P}(Y_j = 1 \mid C = c) = \text{logit}^{-1}(b_j + s a_j z_c)$, where b_j and a_j differ by item: item quality and discrimination are deliberately nonexchangeable. Truth uses finite-mixture analytic moments with an audited bracketed root for the target H .

CLDM (controlled local item dependence). A global latent class plus block-local Rademacher factors, $\mathbb{P}(Y_j = 1 \mid L, U) = q_j + \lambda d_j(L - \pi) + \eta \sum_k \ell_{jk} U_k$: local residual dependence in item blocks that violates the auxiliary conditional-independence story while leaving the fixed-scale estimand well defined (the family’s registered claim scope is “in scope for fixed H ; auxiliary conditional independence violated”). Moderate-dependence, separated-denominator cells remain in the primary regular lane; strong-dependence and boundary cells are stress.

D.2. Subfamily Structure and Partition. Each family contains structurally distinct subfamilies used as whole units in the train/validation/test partition (training: SF01; held-out validation: SF02; the originally scheduled untouched test set: SF03; the fresh confirmatory set generated for the restart: SF04; see [Appendix E](#)). Structural profiles differ within family: for example, LMCI’s subfamilies use an ordered location-slope fan, a lower-margin skew with reversed slope fan, and a zigzag location-alternating quality profile, so that no tuning decision could exploit a single margin geometry. The confirmatory SF04 subfamilies were freshly parameterized variants never touched by any earlier phase.

D.3. The 32 Confirmatory Cells. The design grid crosses $N \in \{60, 120, 250, 500\}$, $J \in \{3, 5, 10, 20\}$, and population targets $H \in \{.150, .250, .280, .295, .300, .305, .320, .450\}$ in eight (N, J, H) combinations shared by all four families ([Figure 2b](#)). Regularity strata were assigned by frozen rules on margins, margin-order gaps, pair denominators, and the normalized scale denominator: 22 cells are primary regular and 10 are stress (three low- N /high- J ; two near-zero pair-denominator; one combined pair-denominator/dependence; one near-zero scale-denominator; three strong local dependence). Regular cells received 4,000 scheduled replications, stress cells 2,000, with $B = 2,000$ BB draws each.

D.4. Truth Oracle Audit. Every cell’s population coefficient was locked before launch with an independent recomputation: across the 32 SF04 cells the maximum absolute truth residual (solver value versus independent re-evaluation) was 6.1×10^{-11} ,

and the launch-locked truth values were canonical for all downstream coverage judgments. During post-processing, a serialization audit compared worker-recomputed truth against launch-locked truth over all 648,000 endpoint rows; the maximum absolute discrepancy was 3.9×10^{-16} (binary round-trip noise, about 2,500 times smaller than the pre-fixed worker tolerance), and the launch-locked values were used exactly.

Appendix E. Development, Selection, and Governance Chronology

E.1. Candidate Library and Training. Before any performance outcome existed, the development phase froze (a) a candidate library of interval constructions (the raw equal-tail BB interval; its basic, studentized, and BCa variants; the NPB percentile interval; the analytic Wald interval; and midpoint-preserving width-corrected versions of the BB constructions) and (b) the calibration grid $\gamma \in \{0, .25, .5, .75, 1, 1.5, 2, 2.5\}$. Training ran on the SF01 subfamilies of the four families with prespecified Monte Carlo budgets, and produced, per candidate, the same unconditional-coverage and conditional-length summaries later used in validation. Training outcomes pruned the library to candidates meeting the frozen coverage and definedness screens.

E.2. Held-Out Selection. Validation ran once on the disjoint SF02 subfamilies. The frozen eligibility criteria (family-level coverage within $[.925, .975]$, catastrophic cell-level protection, definedness, and efficiency screens) left exactly two eligible primary candidates:

- **P7-FS-WIDTH-RAW-G025-v1:** the raw equal-tail BB interval with width calibration $\gamma = .25$;
- **P7-FS-WIDTH-BASIC-G100-v1:** the basic (reflected) BB interval with width calibration $\gamma = 1.0$.

The prespecified selection rule (minimum equal-family, equal-cell conditional mean length, with a 2% relative-tie threshold) selected the former: its mean length was 7.36% shorter than the alternative’s, well beyond the tie threshold, so the tie rule was not invoked. The machine-readable decision record (**phase08-primary-interval-decision-v1.json**) certifies that no acceptance threshold changed and no parameter was modified after validation access.

E.3. Public Semantics Lock. Before any confirmatory data were generated, the complete public method was serialized (**phase08-public-method-semantics-lock-v1.json**): the point estimator; the base interval algorithm (type-8 quantiles of $B = 2,000$ common Dirichlet draws); the exact width transform $1 + .25/\sqrt{n}$ with

no clipping; the inherited-failure policy with undefined intervals counted as unconditional misses; the prohibition of per-replication method cascades; the reporting contract (regular domain primary, stress cells separate); and the deferral of every threshold-evidence quantity (no posterior threshold probability, no threshold score, no rescue role). The lock also fixed the confirmatory acceptance gates before the test.

E.4. The Terminated First Confirmatory Attempt. The originally scheduled untouched test (the SF03 subfamilies) completed its compute stage (12 of 12 workers), but its consolidation stage never produced a scientific quantity, and the two steps involved are exactly documented. First, the original postlaunch consolidator ingested the endpoint CSV and was halted by its own strict parser (a decimal seed token failed the canonical C99 integer check) after the scientific file had been opened but before any metric, summary, gate, or report was computed; the only projection ever taken from SF03 was an outcome-blind audit of 324,000 seed tokens, with no other scientific column materialized. A representation-only repair (bridging the seed columns to canonical integers) was authorized for exactly one further invocation. Second, that single repaired invocation was consumed by an orchestration error: the controller passed the wrong manifest file to a hash-verifying guard, which failed closed before the protected output root was even opened, so that zero rows were read on that attempt. Because the frozen contract prohibited retries, the gate for the attempt was recorded as withheld/not evaluated rather than passed or failed, and the SF03 output was sealed with no metric, summary, or interpretation ever derived from it. The confirmatory results in the main text therefore come from a minimal-change restart on freshly generated SF04 subfamilies with new seeds, under the identical locked method and gates. The restart’s own authority, transfer, and validation chain is archived with SHA-256 digests in the replication package, including a byte-exact raw transfer from the compute environment and two documented, named infrastructure-representation repairs executed before any gate evaluation: a receipt-hash lookup repair (worker receipt hashes had been serialized as an unnamed JSON array; the finalizer was moved to positional binding, with all 12 hashes and all 36 raw-file hashes matching) and a truth-value serialization repair (worker-recorded `truth_h` values were not binary-identical to the launch-locked values; the comparison was moved to canonical serialization). In both cases the finalizer had loaded raw endpoint rows and then failed closed before any coverage, efficiency, or gate quantity was computed, and the

raw rows themselves were never modified. No scientific parameter, method, or acceptance rule changed at any point in this history, and no outcome from the sealed SF03 run informed anything downstream.

E.5. Independent Validations. Each stage carried its own independent check: the theory stage’s derivative and contraction oracles (1,561 checks; [Appendix B](#)); the selection stage’s independent recomputation of the decision; and the confirmatory stage’s independent validator, which re-read all 648,000 stored endpoint rows, recomputed every cell, family, eligibility, and efficiency table, re-evaluated all nine gate criteria, and matched the primary analysis exactly (14 of 14 checks). The final gate decision, a pass with the calibrated interval as the selected primary and no reselection performed, is recorded in `interval-gate-result-v1.json` and `result-lock-v1.json`.

Appendix F. Complete Confirmatory Results

F.1. Gate Criteria and Same-Gate Eligibility. The nine blocking criteria, all passed: (1) integrity (hash-verified reproduction of the frozen evaluation chain, positional worker-receipt binding, and launch-locked truth); (2) family-level unconditional coverage within $[.925, .975]$ in all four families; (3) catastrophic protection (no regular cell below the frozen cell-level lower bound); (4) zero tolerated numerical failures; (5) definedness in the regular lane; (6, 7) two length-efficiency criteria comparing the selected method with same-gate-eligible comparators; and (8, 9) two stress-lane criteria (exact machine-readable definitions in `gate_criteria-v1.csv` in the replication archive). Same-gate eligibility, which requires passing the coverage, catastrophic, numerical-failure, and definedness screens simultaneously, held for exactly three methods: the calibrated interval, the raw equal-tail BB interval, and the NPB percentile interval. The basic BB failed the coverage band (CLDM family macro coverage $.9201 < .925$) and the catastrophic screen; the studentized BB, BCa, and delta-method Wald intervals failed coverage, catastrophic, and definedness screens.

F.2. Cell-Level Tables and Figures. [Tables F.1](#) and [F.2](#) list all 22 primary regular cells for the calibrated and raw intervals: scheduled replications, undefined counts (zero throughout), unconditional coverage, and conditional mean length. [Figure F.1](#) shows all seven methods’ cell-level coverage in one matrix (the columnwise loss of definedness of the derivative-based methods at $J = 10$ and especially $J = 20$ is the failure mode discussed in the main text), and [Figure F.2](#) compares the three same-gate methods’ mean lengths cell by cell. The calibrated-versus-NPB advantage is a

family-level statement: in all four family aggregates the calibrated interval is 1.5%–2.5% shorter at equal or better coverage, but this does not hold cell by cell (the calibrated interval is shorter in 5 and at least as well covered in 18 of the 22 regular cells, with both simultaneously in 2), and no cellwise dominance is claimed.

TABLE F.1. Calibrated BB Interval: All 22 Primary Regular Cells

Family	N	J	Target H	Scheduled	Undefined	Coverage	Length
LMCI	60	3	.150	4,000	0	.9435	0.6626
LMCI	120	5	.250	4,000	0	.9427	0.2468
LMCI	250	5	.295	4,000	0	.9507	0.1717
LMCI	250	10	.280	4,000	0	.9485	0.1108
LMCI	500	10	.305	4,000	0	.9500	0.0793
LMCI	500	20	.450	4,000	0	.9593	0.0678
HMTC	60	3	.150	4,000	0	.9420	0.7648
HMTC	120	5	.250	4,000	0	.9487	0.2719
HMTC	250	5	.295	4,000	0	.9567	0.1930
HMTC	250	10	.280	4,000	0	.9490	0.1087
HMTC	500	10	.305	4,000	0	.9550	0.0780
HMTC	500	20	.450	4,000	0	.9523	0.0546
AQNE	60	3	.150	4,000	0	.9447	0.6399
AQNE	120	5	.250	4,000	0	.9503	0.2468
AQNE	250	5	.295	4,000	0	.9493	0.1742
AQNE	250	10	.280	4,000	0	.9513	0.1074
AQNE	500	10	.305	4,000	0	.9465	0.0775
AQNE	500	20	.450	4,000	0	.9483	0.0640
CLDM	60	3	.150	4,000	0	.9390	0.7113
CLDM	120	5	.250	4,000	0	.9443	0.2619
CLDM	250	5	.295	4,000	0	.9510	0.1856
CLDM	250	10	.280	4,000	0	.9470	0.1054

TABLE F.2. Raw Equal-Tail BB Interval: All 22 Primary Regular Cells

Family	N	J	Target H	Scheduled	Undefined	Coverage	Length
LMCI	60	3	.150	4,000	0	.9355	0.6419
LMCI	120	5	.250	4,000	0	.9357	0.2413
LMCI	250	5	.295	4,000	0	.9477	0.1690
LMCI	250	10	.280	4,000	0	.9450	0.1091
LMCI	500	10	.305	4,000	0	.9473	0.0785
LMCI	500	20	.450	4,000	0	.9573	0.0671
HMTC	60	3	.150	4,000	0	.9313	0.7409
HMTC	120	5	.250	4,000	0	.9423	0.2658
HMTC	250	5	.295	4,000	0	.9530	0.1900
HMTC	250	10	.280	4,000	0	.9473	0.1070
HMTC	500	10	.305	4,000	0	.9520	0.0772
HMTC	500	20	.450	4,000	0	.9497	0.0540
AQNE	60	3	.150	4,000	0	.9360	0.6199
AQNE	120	5	.250	4,000	0	.9433	0.2413
AQNE	250	5	.295	4,000	0	.9467	0.1714
AQNE	250	10	.280	4,000	0	.9470	0.1057
AQNE	500	10	.305	4,000	0	.9450	0.0767
AQNE	500	20	.450	4,000	0	.9440	0.0633
CLDM	60	3	.150	4,000	0	.9313	0.6890
CLDM	120	5	.250	4,000	0	.9400	0.2561
CLDM	250	5	.295	4,000	0	.9470	0.1827
CLDM	250	10	.280	4,000	0	.9435	0.1037

FIGURE F.1. Cell-level unconditional coverage for all seven interval methods in the 22 primary regular cells. Text labels give exact values; undefined intervals count as misses.

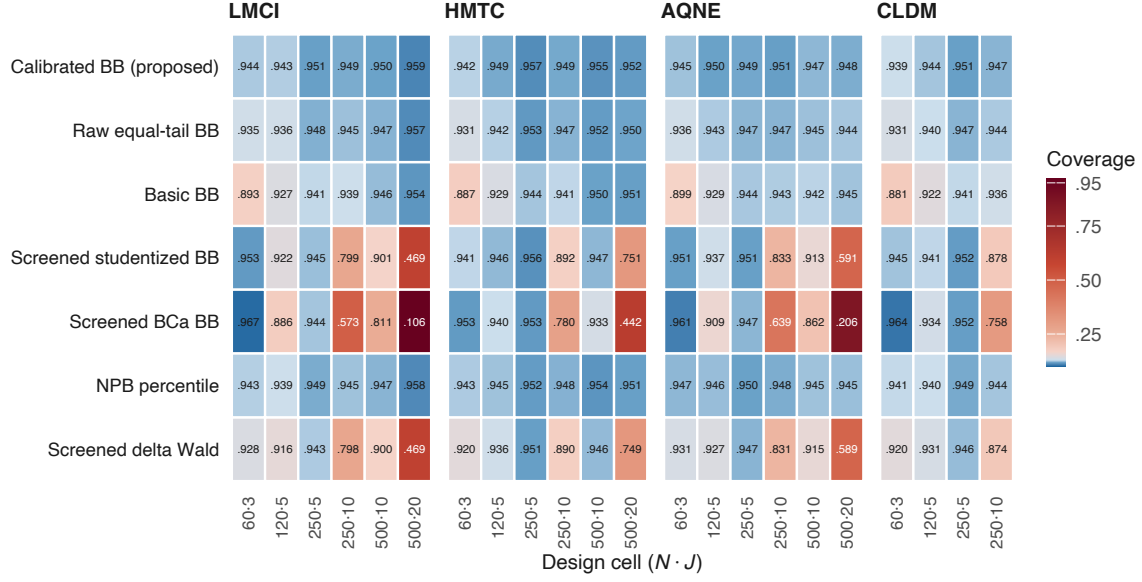
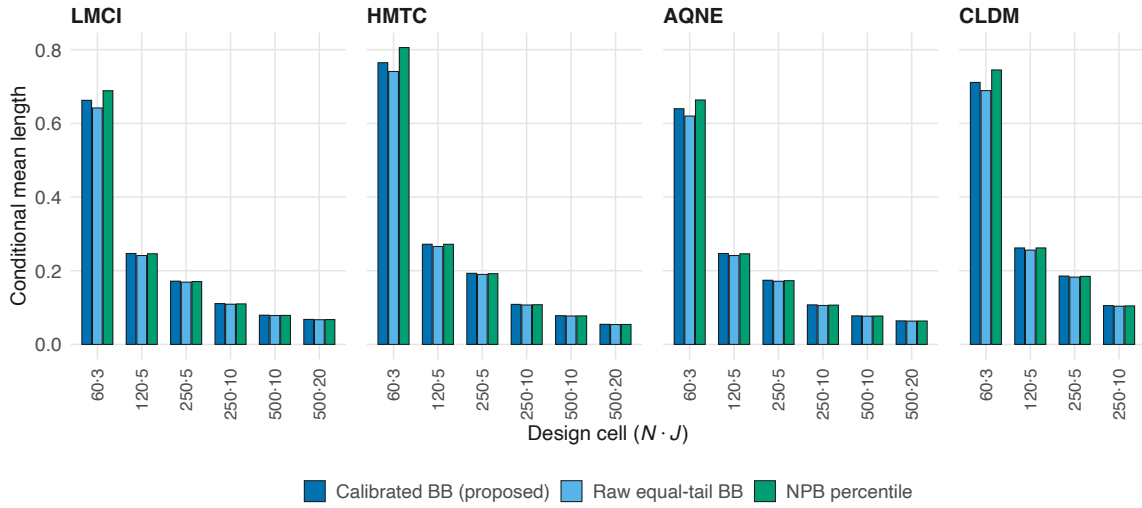


FIGURE F.2. Conditional mean interval length for the three same-gate-eligible methods across the 22 primary regular cells.



F.3. Stress-Lane Detail. Table F.3 reports every method in every stress cell: defined counts over scheduled replications, unconditional coverage, and conditional mean length (dash when no interval was defined). Three regularities: (a) in the low- N /high- J lane the SE- and jackknife-based methods lose definedness almost completely while the weighting/resampling methods hold coverage near nominal; (b) under strong local dependence the BB family’s unconditional coverage remains between .93 and .95; (c) in the near-zero scale-denominator cell every method’s intervals become undefined (at most 1 defined replication of 2,000) and unconditional coverage is zero across the board: the estimand itself approaches non-definability, and the typed semantics report it rather than repair it.

TABLE F.3. Stress-Diagnostic Cells: Definedness, Coverage, and Length for All Seven Methods

Cell (stress lane)	Method	Defined	Coverage	Length
LMCI 60·20 (low n high j)	Calibrated BB (proposed)	2000/2,000	.9445	0.1829
	Raw equal-tail BB	2000/2,000	.9370	0.1771
	Basic BB	2000/2,000	.9295	0.1771
	Screened studentized BB	2/2,000	.0010	0.1845
	Screened BCa BB	0/2,000	.0000	—
	NPB percentile	2000/2,000	.9455	0.1831
	Screened delta Wald	2/2,000	.0005	0.1849
HMTC 60·20 (low n high j)	Calibrated BB (proposed)	2000/2,000	.9420	0.1534
	Raw equal-tail BB	2000/2,000	.9325	0.1486
	Basic BB	2000/2,000	.9360	0.1486
	Screened studentized BB	24/2,000	.0115	0.1438
	Screened BCa BB	0/2,000	.0000	—
	NPB percentile	2000/2,000	.9385	0.1533
	Screened delta Wald	24/2,000	.0115	0.1461
AQNE 60·20 (low n high j)	Calibrated BB (proposed)	2000/2,000	.9470	0.1679
	Raw equal-tail BB	2000/2,000	.9440	0.1627
	Basic BB	2000/2,000	.9440	0.1627
	Screened studentized BB	4/2,000	.0015	0.1577
	Screened BCa BB	0/2,000	.0000	—
	NPB percentile	2000/2,000	.9480	0.1679
	Screened delta Wald	4/2,000	.0015	0.1598
LMCI 120·3 (pair denominator boundary)	Calibrated BB (proposed)	2000/2,000	.9450	0.6000
	Raw equal-tail BB	2000/2,000	.9390	0.5866

Cell (stress lane)	Method	Defined	Coverage	Length
AQNE 120·3 (pair denominator boundary)	Basic BB	2000/2,000	.9120	0.5866
	Screened studentized BB	0/2,000	.0000	—
	Screened BCa BB	0/2,000	.0000	—
	NPB percentile	2000/2,000	.9400	0.6137
	Screened delta Wald	0/2,000	.0000	—
	Calibrated BB (proposed)	2000/2,000	.9445	0.6124
	Raw equal-tail BB	2000/2,000	.9395	0.5988
	Basic BB	2000/2,000	.9055	0.5988
	Screened studentized BB	0/2,000	.0000	—
	Screened BCa BB	0/2,000	.0000	—
CLDM 120·3 (pair denominator boundary and strong local dependence)	NPB percentile	2000/2,000	.9445	0.6341
	Screened delta Wald	0/2,000	.0000	—
	Calibrated BB (proposed)	2000/2,000	.9105	1.2562
	Raw equal-tail BB	2000/2,000	.9075	1.2282
	Basic BB	2000/2,000	.7810	1.2282
	Screened studentized BB	1/2,000	.0000	1.2327
	Screened BCa BB	0/2,000	.0000	—
	NPB percentile	2000/2,000	.9115	1.3259
	Screened delta Wald	1/2,000	.0000	1.2801
	Calibrated BB (proposed)	1/2,000	.0000	0.0000
HMTC 120·3 (scale denominator boundary)	Raw equal-tail BB	1/2,000	.0000	0.0000
	Basic BB	1/2,000	.0000	0.0000
	Screened studentized BB	0/2,000	.0000	—
	Screened BCa BB	0/2,000	.0000	—
	NPB percentile	0/2,000	.0000	—
	Screened delta Wald	0/2,000	.0000	—
	Calibrated BB (proposed)	2000/2,000	.9350	0.1499
	Raw equal-tail BB	2000/2,000	.9245	0.1452
	Basic BB	2000/2,000	.9385	0.1452
	Screened studentized BB	28/2,000	.0135	0.1509
CLDM 60·20 (strong local dependence)	Screened BCa BB	0/2,000	.0000	—
	NPB percentile	2000/2,000	.9365	0.1495
	Screened delta Wald	28/2,000	.0135	0.1532
	Calibrated BB (proposed)	2000/2,000	.9545	0.0757
	Raw equal-tail BB	2000/2,000	.9510	0.0749
	Basic BB	2000/2,000	.9500	0.0749
	Screened studentized BB	1988/2,000	.9445	0.0752
	Calibrated BB (proposed)	2000/2,000	.9545	0.0757
	Raw equal-tail BB	2000/2,000	.9510	0.0749
	Basic BB	2000/2,000	.9500	0.0749
CLDM 500·10 (strong local dependence)	Screened studentized BB	1988/2,000	.9445	0.0752
	Calibrated BB (proposed)	2000/2,000	.9545	0.0757
	Raw equal-tail BB	2000/2,000	.9510	0.0749
	Basic BB	2000/2,000	.9500	0.0749

Cell (stress lane)	Method	Defined	Coverage	Length
CLDM 500·20 (strong local dependence)	Screened BCa BB	1949/2,000	.9270	0.0750
	NPB percentile	2000/2,000	.9495	0.0752
	Screened delta Wald	1988/2,000	.9445	0.0751
	Calibrated BB	2000/2,000	.9565	0.0532
	(proposed)			
	Raw equal-tail BB	2000/2,000	.9530	0.0526
	Basic BB	2000/2,000	.9495	0.0526
	Screened studentized BB	1527/2,000	.7280	0.0525
	Screened BCa BB	883/2,000	.4240	0.0526
	NPB percentile	2000/2,000	.9535	0.0528
	Screened delta Wald	1527/2,000	.7290	0.0527

F.4. Monte Carlo Precision. With 4,000 scheduled replications per regular cell, the MCSE of a cell-level coverage proportion near .95 is approximately .0035; family-level macro coverage MCSEs range from .0014 to .0022 across methods and families (parenthesized throughout [Table 1](#) of the main text). Calibrated-versus-raw coverage differences are paired contrasts computed on identical replications and identical weight draws, so their Monte Carlo error is far smaller than the two marginal MCSEs suggest.

Appendix G. Scientific Reasoning Scale Application: Details

G.1. Source, License, and Provenance. The data are the public Italian administration of the 11-item Scientific Reasoning Scale (Drummond & Fischhoff, 2017): “Psychometric properties of the Scientific Reasoning Scale: Application to the Italian context” (Caliciuri & Lanz, 2024), OSF project [jk9dp](#), CC-BY 4.0. Collection ran 2024-03-25 through 2024-05-31 via three Qualtrics forms with different question orders; recruitment was snowball sampling of Italian adults (18+). Of 624 received responses, 606 were consent-eligible and 337 answered all 11 items. Source files (analysis data, readme, codebook) were verified against their OSF SHA-256 digests, mirrored read-only into a checksummed local warehouse, and consumed without modification.

G.2. Outcome Firewall. The eligibility decision (this dataset, the complete-case frame, and the full fixed 11-item set) was locked before the source’s own psychometric report was read and before any coefficient, interval, or diagnostic was computed. In particular: the original authors’ reduced (six-item) subset was not used; no item was reversed, deleted, or selected at any point; and none of the source’s CTT or IRT results

entered any decision. The firewall flags are recorded in the archived verification file (`ppsr-s-osf-source-verification-v1.yml`).

G.3. Analysis Constants and Diagnostics. The frozen algorithm ran with $B = 50,000$ draw-major weight draws, seed 2026072611, type-8 quantiles, $\gamma = .25$, no clipping. All 50,000 draws were defined; the smallest weighted scale denominator over the draws was 8.30 (nowhere near the degeneracy boundary), and the smallest empirical pair denominator was .10, well separated from zero. Batch-based Monte Carlo standard errors of the calibrated endpoints were .00017 (lower) and .00019 (upper), both far below the frozen .0015 limit. Three implementations of the point estimate (direct pair aggregation, the vectorized total-score identity used by the BB engine, and `mokken`’s `coefH`) agreed to 3×10^{-15} . Reversing the fixed item order under fixed weights left the point estimate and both calibrated endpoints unchanged to numerical precision (10^{-15} level).

G.4. Item and Pair Descriptives. [Table G.1](#) lists the fixed items’ keyed proportions and descriptive H_i values with `mokken` standard errors (descriptive only: no item-level calibrated interval exists, and no selection rule was applied). Five of 55 pair coefficients are negative (minimum $-.067$); the maximum pair coefficient is .644 (SRS8–SRS10).

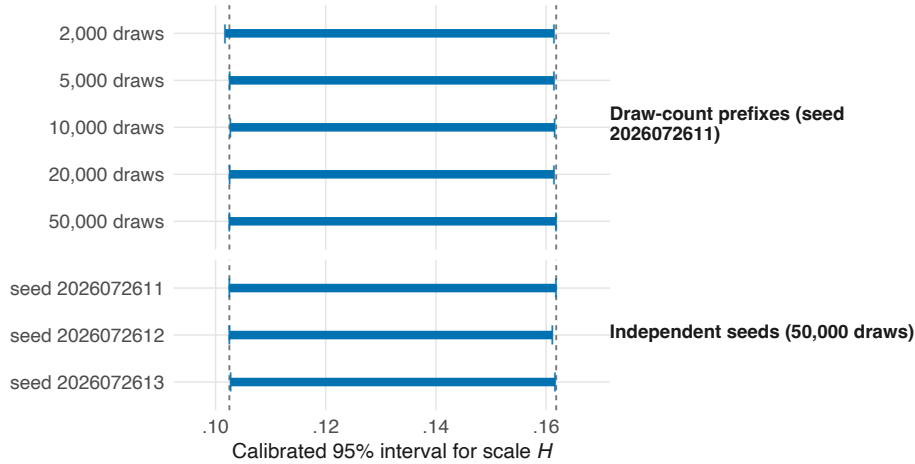
TABLE G.1. Descriptive Item Statistics for the Fixed 11-Item Scientific Reasoning Scale

Item	p_1	H_i	<code>mokken</code> SE(H_i)
SRS1	.570	.053	.031
SRS2	.691	.166	.026
SRS3	.712	.061	.029
SRS4	.561	.117	.030
SRS5	.662	.078	.028
SRS6	.709	.187	.026
SRS7	.668	.081	.028
SRS8	.754	.228	.027
SRS9	.653	.159	.027
SRS10	.407	.229	.036
SRS11	.490	.110	.033

G.5. Sensitivity. [Figure G.1](#) displays the calibrated interval across three independent 50,000-draw seeds and across draw-count prefixes of the main stream. Across

seeds the lower endpoint varies by .00026 and the upper endpoint by .00071; across prefixes from 2,000 to 50,000 draws the interval's relation to the descriptive .30 reference never changes. The BB mean and median exceed the plug-in point estimate by .0008 and .0005, respectively (the small nonlinear center shift whose order [Equation \(B.7\)](#) anticipates), and were reported, not used to recenter the interval.

FIGURE G.1. Stability of the calibrated 95% interval for the Scientific Reasoning Scale across independent seeds and draw-count prefixes. Dotted verticals mark the main-analysis endpoints.



Appendix H. Software and Reproducibility

H.1. **The bayesmokken Package.** The public entry point is `bb_mokken_h()`:

```
library(bayesmokken)
fit <- bb_mokken_h(data, draws = 2000, seed = 1,
                  interval = "calibrated", level = 0.95)
fit$interval      # method id, point, raw + calibrated endpoints,
                  # gamma, width multiplier, draw counts, typed status
```

Design decisions keep this call identical to the validated method: `interval` accepts only "calibrated" and "equal_tail"; the confidence level is fixed at .95 (the only level with calibration evidence); γ is not a user argument; no threshold probabilities, posterior summaries, or item-selection utilities are exposed; and undefined states are returned as typed statuses, never silently repaired. A deprecated alias (`bayes_mokken_h()`) forwards with a warning for early adopters. The application data are included as the `srs` dataset. At the application gate for the binary method the package

passed 665 unit tests (0 failures, 0 warnings, 0 skips) covering the deterministic core, the weight-stream contract (identical draws for any batch size and worker count at a fixed seed), quantile and calibration algebra, typed failure paths, and print/summary methods; R CMD `check` completed with status OK; and the saved 50,000 Scientific Reasoning Scale draws re-entered through the installed package reproduced the frozen interval endpoints to 1.1×10^{-15} .

H.2. Scope of the Installed Package Versus This Article. The installed package is broader than the procedure evaluated in this article, and the difference should be stated explicitly so that availability is not mistaken for validation. Alongside the binary path above, `bb_mokken_h()` accepts `item_type`, `supports`, `cluster`, `cluster_target`, and `weight_law`, which reach a separately developed procedure for finite-support *ordinal* scales with *clustered* respondents: a different estimand (comonotone-normalized weighted multicategory H), a two-stage hierarchical Dirichlet weight law with respondent-weighted or equal-cluster targets, and its own tuning constant, of opposite sign to the one studied here. That procedure has its own development, held-out validation, fresh-confirmation, and application chain, reported in companion work; nothing in the present theory, simulation, or application bears on it. The package enforces the boundary rather than guessing: dichotomous data with a cluster argument are refused on every route, ordinal data with the independent-respondent law are refused, and the clustered path requires its estimand target and item supports to be declared explicitly rather than defaulted. For reproducing this article, the binary call and its method identifier (P7-FS-WIDTH-RAW-G025-v1) are the complete contract.

H.3. Release History Relevant to This Article. The version evaluated at the application gate was 0.0.0.9000. Two corrections made after the gate are disclosed for completeness, neither affecting any number in this article: an external review found (a) that the deterministic per-pair *ordinal* transport kernel, which the binary path reported here does not use, could drop a valid positive mass at or below 10^{-12} , and (b) that a fractional seed passed to the draw-stream constructor was silently truncated rather than rejected. Version 0.0.0.9001 fixed both and added adversarial tests. The public release at the time of writing is 0.9.0, which adds a documented user-facing surface (data screening, result extraction, diagnostics, reporting, and figures) while carrying the estimation engine over unchanged: the release’s parity tests compare the frozen 0.0.0.9001 numerical receipts field by field at zero tolerance, including all draw

arrays, and the calibration constants and the 95% level remain unreachable from the public interface. The call forms printed in this article run unchanged.

H.4. Repositories.

- Package:
<https://github.com/joonho112/bayesmokken>
- Replication archive:
<https://github.com/joonho112/bayesmokken-scalability-replication>

The replication archive rebuilds every figure and table of this article and its supplement from the locked artifacts under hash manifests; the archived raw outputs make re-running the original computation unnecessary, and every random stream is seeded and draw-major, so results are independent of batch size and worker count.